Overcoming Gridlock 2.0

Solving Power Bottlenecks to Drive AI & Energy Security

Citi Research

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Executive Summary

Slowly, then suddenly. Demand for power, infrastructure, and generation equipment continues to rise, driven primarily by the U.S.-focused AI data center buildout. The Middle East conflict has sharpened focus on energy security. As countries seek to reduce import dependence and limit exposure to price spikes, the reaccelerated adoption of renewables, batteries, and electric vehicles (EVs) will also raise power equipment demand. However, grid expansion appears unable to keep pace with strong demand and supply growth, constrained by equipment supply-chain challenges and well-meaning permitting frameworks in many regions. Thus, speed-to-power is driving a shift from centralized networks toward more distributed systems that bypass grid-connection delays. "Bring-your-own-generation" (BYOG), particularly U.S. natural gas–based supply, continues to gain traction. This report examines global power markets and explores solutions to ease grid constraints and accelerate AI deployment and energy security transitions.

AI is transformational and needs substantial power capex. Some caution that certain aspects of AI look like a bubble. Our stance is that power is still very much needed as the Jevons Paradox (Fortune, April 28) takes hold, and we see the related capex being utilized in the future.

Moreover, energy security is now top-of-mind for many countries. This is the top concern in the latest International Energy Agency report on Southeast Asia. Unlike imported oil and natural gas, renewable energy sources are typically domestically located. But relying on imported equipment for renewables introduces its own set of national security considerations. Expanding the share of electric vehicles (EVs), which are powered largely by domestically generated power, helps to cut oil imports. Even if AI power growth were to slow, renewables, EVs and distributed generation still require more of certain types of infrastructure equipment.

And yet, the power grids needed to facilitate the AI and energy security transformations are typically enormous and slow to improve after decades of electrification and economic growth. Speed-to-power is pushing the evolution of the grid. Our prior study, Overcoming Gridlock: Powering Our Future (May 2024), had explored solutions, but power demand growth has materialized much stronger than expected. **The demand surge in power infrastructure equipment has strained supply chains and extended the wait time. Grids will still expand, but consumers of power cannot wait.**

Thus, speed-to-power concerns drive demand for power generation equipment: from more efficient combined cycle gas turbines; to smaller, less efficient reciprocating engines, aeroderivative turbines and fuel cells; to the use of renewables and energy storage systems. Much of the above-listed generation equipment demanded in the U.S. requires natural gas supply, **which could lead to a further boom in the U.S. midstream space.**

This report draws on experts across the firm to examine this megatrend — Research (global commodities, utilities, industrials, materials and U.S. midstream), and Q&As with Banking and Wealth — for a holistic study.

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A Conversation with Citi Banking

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There is strong demand for power equipment, driven by strong growth in data center power demand and the reacceleration of renewables and electric vehicles in parts of the world, partly due to energy security needs.

> *What do you see as key supply-chain bottlenecks in the equipment space?*

The supply chain of Transmission & Distribution and Gas Generation is most constrained. Gas Turbines are costing 2-3x what they used to cost and there is a shortage of substations/transformers and switchgear equipment.

> *Do you think the market underestimates the scale and duration of these supply-chain bottlenecks?*

We believe the market has not understood the real issues with the supply chain and there is general ambiguity around the whole supply chain of actual backlog/delivery timelines.

With power grid improvements not necessarily keeping pace with power demand, more project developers opt for "bring your own generation" (BYOG)[1]. Yet, order books of gas turbines are very strong now that newer-generation ideas have emerged.

> *As OEMs have historically been cautious about adding too much capacity, are turbine and engine manufacturers and their suppliers confident in AI-driven power demand growth that justifies capacity expansion?*

The demand is real, but for adding capacity the OEMs have to take a much longer view for ensuring they meet hurdle rates. As such the decision making is slower, causing buyers to focus on other sources of supply.

> *How are newer distributed generation ideas — say aero-derivative turbines, recip engines or others — looking like substitutes?*

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[1] BYOG refers to arrangements where customers procure or contract their own power supply, which may be onsite or offsite (e.g., a data center securing dedicated generation via a power purchase agreement, or PPA).

They are re-purposing the gensets/engines from original use in oil fields/aviation to provide behind-the-meter (BTM) solutions to data centers.[2] The use-case value is strong, but supply is limited and the solution is more nuanced and complicated.

Some data center developments are delayed or canceled. What does a typical development cycle of data centers look like and what are key aspects that affect their timing?

A typical data center build-out is two years, while the power needed for these if new generation is required could take 3-5 years, leaving a period of gap. The lack of power and complications in interconnect is pushing developers to prioritize places where interconnect and power supply has more visibility.

***Cathy Shepherd** is Citi's Global Head of Energy Corporate Banking. She is responsible for strategy and capital allocation for Energy clients globally. Cathy has over 20 years of banking experience in the US and Europe with a specialty in working with our Energy and Clean Energy clients across a range of transactions including debt, equity, acquisition finance, derivatives and other credit related products.*

U.S. data centers are increasingly powered by generation equipment that uses natural gas. This has implications for the outlook on gas in the U.S. and globally.

How are energy companies in the U.S. responding to the expected increase in natural gas demand to fuel AI data center growth?

The AI data center boom has become one of the biggest drivers of new electricity demand forecasts, and natural gas is expected to be a key part of the power supply to meet this demand. Energy clients are directly working with developers and hyperscalers to find locations to colocate data centers next to existing natural gas resources and infrastructure, leading to the BYOG trend. In addition, our clients are actively pursuing data center contracts, especially in key regions like Virginia, Texas, Georgia and Ohio to support additional infrastructure investment including new pipeline connections, additional compression and storage infrastructure, etc.

Is data center gas demand large enough and contracted enough to justify new pipeline infrastructure?

The industry is in the early stages of addressing this infrastructure question. If we see growth and corresponding demand rise to levels projected, it's hard to imagine a world where we will not need development of gigawatts of new gas-fired power plants and corresponding natural gas infrastructure to meet this demand. Today the focus is on quick access to markets leveraging existing infrastructure. Developers and hyperscalers are increasingly colocating data centers near existing gas resources, thereby leveraging established infrastructure. This is a key factor behind the significant project concentration in Texas, which benefits from proximity to the Permian Basin.

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[2] BTM refers to onsite generation located on the customer side of the meter (e.g., a data center installing onsite gas generation or solar + storage).

How much could gas for data centers be a short-term, 5-year bridge, or a longer term, 20-year baseload opportunity, or more like a backup/flexibility product?

It's really all of the above. Natural gas plays a multifaceted role in powering the data center expansion. For developers prioritizing speed, it is the essential fuel for "bridge-to-grid" projects, where on-site generation provides baseload power while awaiting grid connection. This has led to a strategy of building on-site generation near existing gas infrastructure. However, this approach is severely hampered by gas turbine backlogs of up to five years. Looking ahead, as the grid expands, natural gas will remain a cornerstone of the baseload power mix alongside nuclear and renewables, ensuring sustained demand growth across all time horizons.

Which geographical areas or shale plays are garnering the most interest?

As power availability is quickly becoming a key constraint to data center growth, developers are bypassing grid constraints by building new power generation directly where the fuel is, leading to a strategic concentration of projects in Texas, near the Permian Basin, and we're increasingly seeing demand grow in the Marcellus-Utica shale region of Pennsylvania and Ohio.

How much can gas-fired power generation serving data centers run during more extreme weather situations, such as cold weather peaks when heating demand would be very high? What are potential mitigating measures?

I think this is less about natural gas and more about need to build hybrid systems that pair grid connections with captive power, battery storage, etc. Developers are acutely focused on continuous baseload power. We are increasingly seeing developers agree to curtail usage during peak hours to secure faster grid connections. Where captive generation is fed by natural gas this adds a layer of security for the power generation. The use of hybrid systems that pair gas with battery storage provides a mitigating measure. Batteries can be dispatched to handle short-term needs, reducing the immediate strain on the grid or on-site gas generation during peak events.

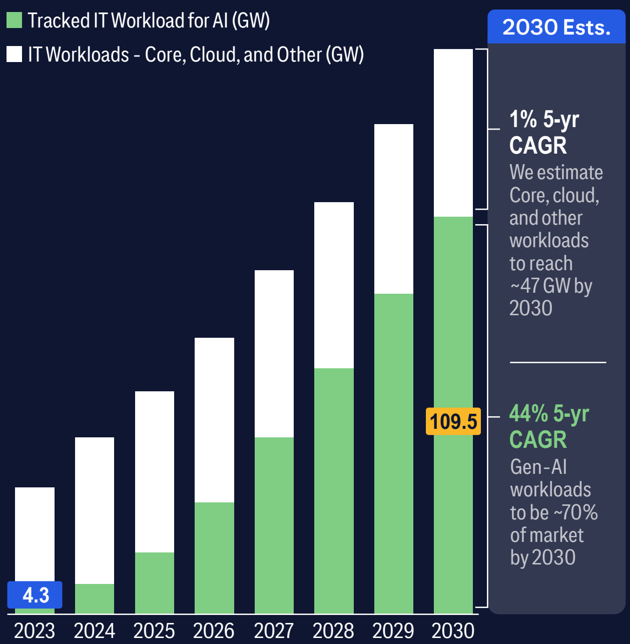
To what degree do you think gas would be used elsewhere globally in power data centers? Or is it more of a U.S. phenomenon?

That will vary by region. In countries and regions with meaningful natural gas resources and infrastructure we would expect trends to be similar to the U.S. For example, in Alberta, Canada where we are already seeing similar colocation trends emerging. For countries that import natural gas, they will take an all-energy approach based on local needs and economics to build out additional power to meet rising AI-driven data center demand growth.

AI & Energy Security

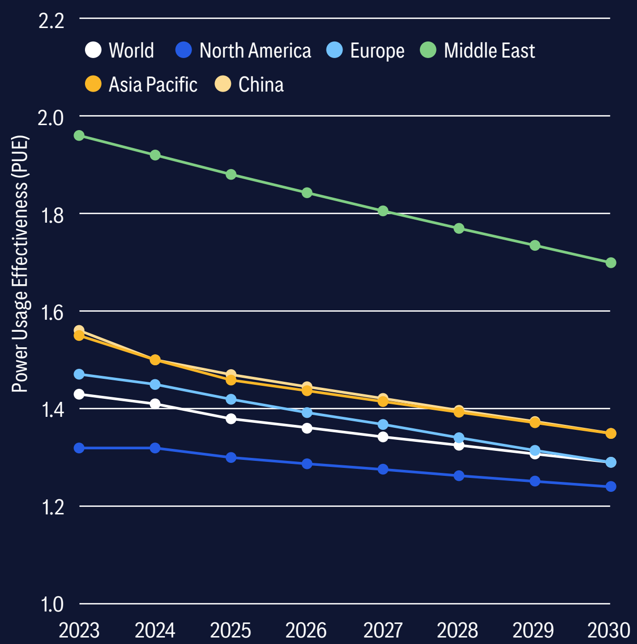
AI Drives Data Center Growth

AI is the dominant source of tracked data center demand increasing form 4.3 GW (~12% share) in 2023 to approximately 110 GW (70% share) by 2030.



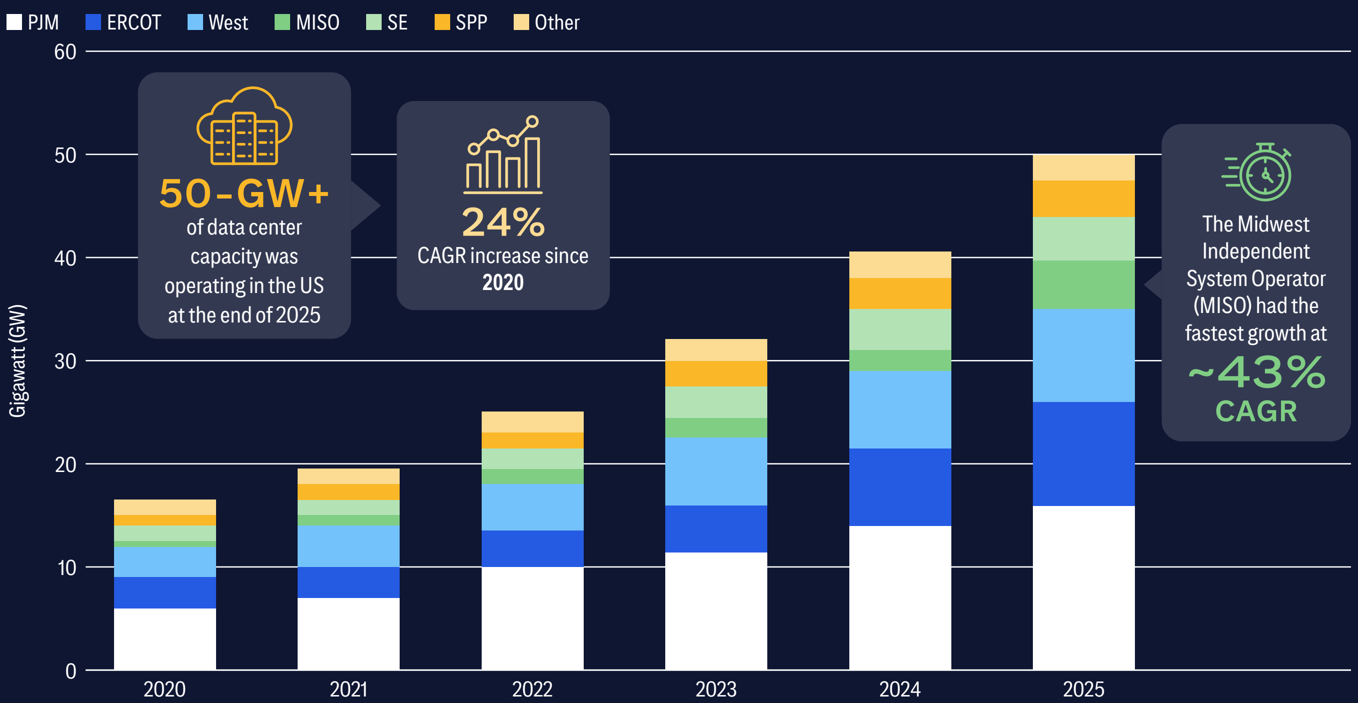
Lower PUE, Higher Computing Efficiency

Declining PUE reflects ongoing efficiency gains that help ease power constraints while supporting AI-driven growth.



Sizeable Growth of In-Service Data Center Capacity

U.S. data center capacity exceeded 50 GW in 2025, growing at a 24% CAGR since 2020. Growth was led by MISO (43% CAGR), with ERCOT, SPP, and SE also experiencing strong expansion as AI-driven demand accelerates.



Sources: Citi Telecommunications Research , Woodmac, FERC

U.S. Power Value Chain: Key Infrastructure Opportunities

Opportunities in the equipment space are vast as numerous industries all need similar electrical hardware. More than 2,500-GW of renewable, storage, and large-load projects are already stalled in grid queues worldwide requiring annual grid investment, expansion of grid supply chains, and workforce capacity.

Meeting electricity demand through 2030 requires **annual** grid investment to rise about **50%** from today's **~$400 billion**

Electrical Equipment Market

- Market expected to grow to $65 billion by 2030
- Critical Component Lead Times = 18-36 months

Generator Step-Up Transformers

- Used in natural gas-fired power plants, solar/wind/storage plants and BYOG facilities
- Demand increased more than 270% since 2019
- Large Unit Lead Times = up to 3- 4 years

Large Power Transformers

- Used in utility substations, data center substations, and transmission networks
- Large Unit Lead Times = 24-48 months

Medium/High-Voltage Switchgear & Breakers

- Used in data center electrical rooms, substations, and grid interconnects
- Unit Lead Times = 12 - 30+ months

Substation Transformers

- Demand up by more than 110% since 2019
- Prices are about 80% higher over the last five years

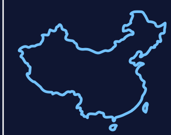
Global EV Demand Reaccelerates

Global EV sales surpassed 20 million units in 2025, rising ~20% year-over-year, led by strong growth in Europe, China and emerging markets.

Europe

- EV Sales rose ~30% in 2025 (4.2 million)
- EV Production rose ~30% to ~3.2 million

China



- EVs accounted for ~55% of new car sales (13 million+)
- Produced ~75% of global EV output (~16 million)

Southeast Asia

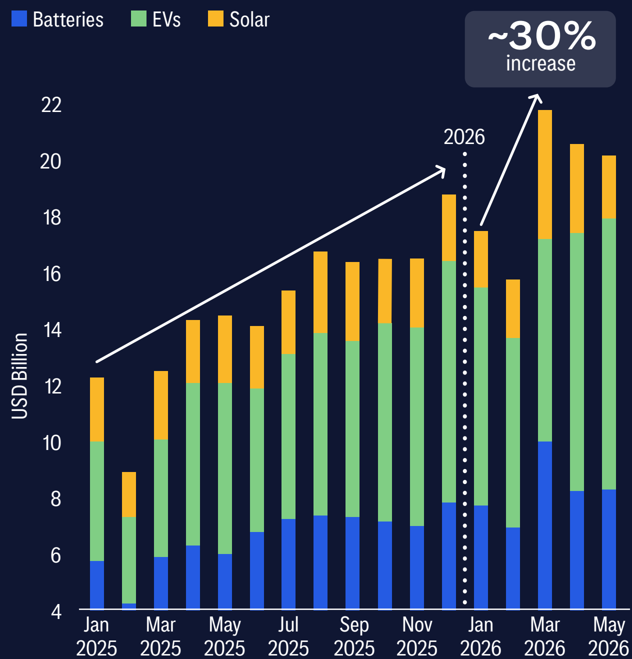
- EVs represented ~ 40% of new car sales in Vietnam
- EV penetration approached 25% in Thailand

Middle East

- EV sales reached ~75K units in 2025
- ~40% growth y/y

Global Demand for Energy Security Boosts Chinese Clean-Tech

Rising energy security concerns are accelerating investment in renewables and energy storage, driving record demand for Chinese clean-tech exports.



Sources: Citi Research, IEA, Woodmac, Reuters

Overcoming Gridlock

The drivers: AI and energy security

Commodities Strategy Views

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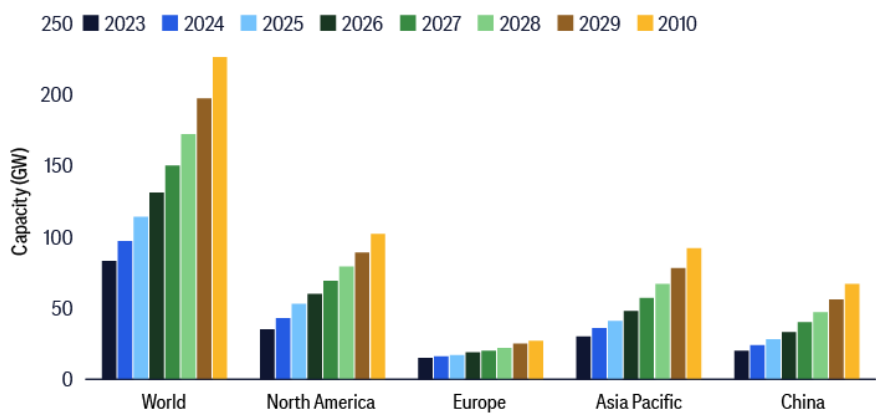
1. What the AI market is telling us about power scarcity

The AI market signals that power access is a revenue and cost constraint in addition to being an infrastructure concern. Frontier model providers continue to strike bespoke compute-and-power deals with infrastructure owners, while hyperscalers are using their balance sheets to reserve capacity ahead of demand. Major model providers are making significant capacity-reservation moves in a scarce power-and-compute environment with the goal of substantially increasing capacity.

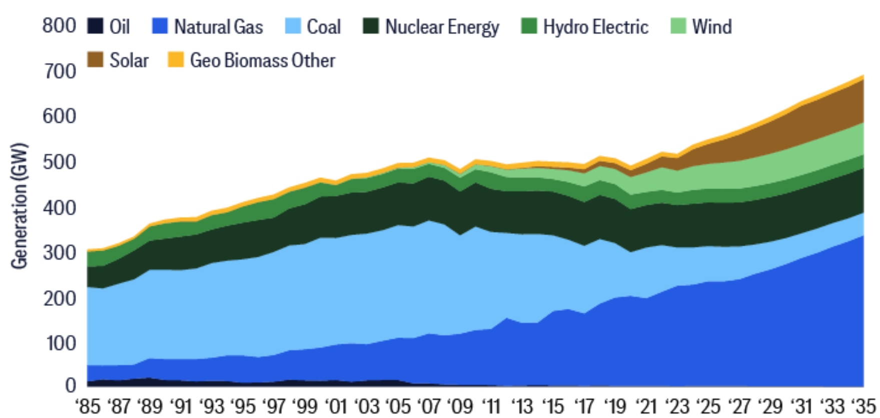
Figure 1. Global data center capacity to rise non-linearly, particularly as demand climbs and deliveries of power infrastructure and generation equipment rise in the years ahead



Source: Citi Research, IEA

More broadly, data center electricity demand rose 17% in 2025, with AI-focused facilities growing even faster, and that tightening deployment bottlenecks are driving a "scramble for solutions". ([IEA](), April 16) U.S. data centers are increasingly powered by natural gas-fired generation.

Figure 2. Rising U.S. power demand, driven primarily by data centers, will increasingly be met by natural gas-fired generation



Source: Citi Research, EIA

However, constraints cover the full stack of energization — transformers, switchgear, turbines, fuel cells, substation work, and skilled labor. That is why the market is increasingly bifurcating between those who can secure physical capacity and those who must buy access from them. This strengthens the hand of hyperscalers and infrastructure owners relative to model labs that do not control cloud, chips, sites, or power procurement.

Improving energy efficiency is a key focus, through both better performance per watt and power usage effectiveness (PUE). Better performance per watt takes place at the chip and software level through improvements in liquid cooling designs, re-architecting software and newer silicon generation to increase the output per watt. Put differently, PUE improvements occur at the facility level before electricity even gets to the servers.

As AI costs rise, we note a shift under way in select use cases toward open-weights (partially open) models, reflecting their lower inference costs relative to frontier closed-source models.[3] (Toms Hardware, May 23) However, lower inference costs would drive higher overall compute demand, increasing data center utilization — consistent with Jevons Paradox, where falling costs lead to greater usage. (Bloomberg, June 10) Model use in some areas would partly migrate from frontier models such as GPT-5.5 and Opus 4.8 to those such as GLM 5.2 or V4 Pro or their equivalents in the future. The scaling law from Hoffman (2022), which had shown that optimal performance under a fixed compute budget is achieved by scaling parameters and training tokens jointly and approximately linearly[4] (and which has helped justify vast investments in the past), has since been superseded by over-training smaller architecture at the pre-training stage, using various reinforcement learning (RL) techniques at the post-training stage, and more extensively deploying chain-of-thought, Mixture of Experts and other techniques at the inference stage. Yet investments remain high owing to the desire to come out on top in AI and to the proliferation of inference demand.

2. Energy security through energy transition

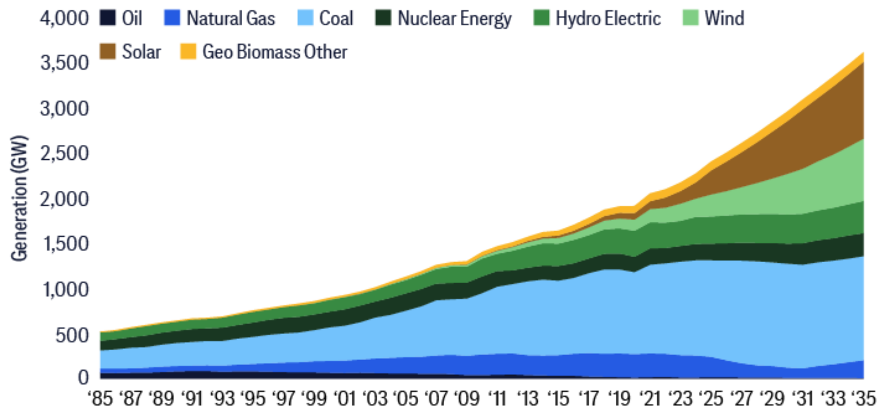
Geopolitical stress around the Middle East, along with price volatility and spikes, reinforces the case for reducing marginal LNG and oil exposure. The IEA's recent "Southeast Asia Energy Outlook 2026" reinforces our view on the region's energy security imperative, further underscoring the critical need to accelerate the deployment of renewables (IEA, 06/16/2026).

To reduce reliance on LNG and oil imports, countries can increase the use of domestic energy sources such as solar, wind, and hydro, alongside batteries, demand response, energy efficiency, and nuclear restarts or life extensions. Electrification — particularly through greater adoption of electric vehicles, which typically rely on domestically generated power — can also contribute. However, constraints in power equipment and infrastructure mean this transition will not happen quickly. Longer-lead solutions include transmission and distribution upgrades (e.g., substations), offshore wind development, large-scale grid storage, new thermal and nuclear plants, and interregional transmission corridors.

[3] See also Artificial Intelligence - Citi's Inference Ahead - Open v. Closed

[4] Hoffmann et al. (2022), *Training Compute-Optimal Large Language Models* (DeepMind), which introduces the "Chinchilla" approach to compute-optimal scaling of model size and training data.

Figure 3. Although total power demand in LNG-importing countries will continue to rise (using Asia+Europe as a proxy below), stronger renewables capacity installations would help limit LNG demand, particularly in Europe and China



Source: Citi Research, BNEF, Energy Institute, IEA

Figure 4. A region-by-region look at the exposure to LNG imports and security pressure, and potential outcomes from substitutions away from LNG into other power generation sources

Region	LNG and security pressure	Grid or equipment constraints	Potential outcome
Europe	High after Russian gas exit and LNG reliance	High grid congestion, extensive permitting process, equipment shortages	More renewables+batteries, but gas capacity still valuable but decline in importance
Japan, Korea, Taiwan	Very high LNG import exposure	Grid and land constraints	Continued nuclear restarts and lift extensions, greater use of batteries, but gas capacity still valuable
Mainland China	Moderate LNG import exposure but high security sensitivity	Strong domestic manufacturing and grid buildout capacity	Turning more to domestic sources: renewables, coal, nuclear and batteries
India	High LNG price sensitivity	Grid and financing constraints, equipment shortages	Accelerated solar and battery deployments; continued reliance on coal
Southeast Asia	High LNG price and import sensitivity	Grid and financing constraints, equipment shortages	Growth in solar and battery deployments; continued use of coal gas and oil-fired generation
Middle East	Export disruption and domestic gas allocation concerns	Strong financing, but cooling and water constraints	More solar+batteries to displace fossil fuel generation to export more fossil fuel
Africa	High LNG price sensitivity	Grid and financing constraints, equipment shortages	Accelerated solar and battery deployments
LatAm	Mixed LNG exposure	Grid and financing constraints, equipment shortages	Continued expansions in solar and wind
US	LNG exporter with abundant gas	Grid constraints, equipment shortages	Some additional demand for US LNG; data center to drive gas demand growth

Source: Citi Research

Germany, India, and ASEAN, among others, are moving in that direction. The German government announced a €8 billion climate program including measures to expand wind power and support EV adoption (DW, March 25). India is accelerating clearances for wind power plants and battery energy storage systems as the Middle East war has brought gas availability issues and price volatility (Reuters, March 30). ASEAN has adopted a 2026–2040 energy cooperation plan targeting renewables to reach 45% of installed electricity capacity by 2030; however, limited grid interconnection remains a key bottleneck to investment and implementation (OPIS "Solar Seen as Energy Security Asset amid Iran Conflict", April 7).

EV demand is also reaccelerating globally outside the U.S., especially in Europe and Asia. EV manufacturing is rising to meet this demand. EV sales momentum picked up with the sales recovery in 2025 and received an extra boost from the Middle East conflict. Global EV sales grew 20% in 2025 to more than 20 million, reaching 25% of global car sales. In Europe, EV sales rose ~30% in 2025 to 4.2 million, or ~28% of new car sales. China sold more than 13 million, making up ~50% of new car sales. In Southeast Asia, EV sales reached 40% of new car sales in Vietnam and nearly a quarter in Thailand. In the Middle East, EV sales reached ~75k in 2025, up ~40% year over year (YoY). In contrast, U.S. EV sales in 1Q 2026 were down ~27% YoY.[5] Subsequently, European new EV registrations rose 34% YoY in April 2026, the first month to fully reflect the oil shock from the Iran war. There have been sharp increases in EV enquiries and orders across Renault, Volvo, Carwow and OLX (Reuters, May 20). In Asia, car buyers have been switching to EVs amid higher fuel costs and, in some places, gasoline shortages (Reuters, April 1).

The sales trend looks durable in Asia and Europe, unlike in the U.S. In Europe, the oil shock is interacting with three structural drivers: lower EV prices, stronger carbon rules, and more affordable Chinese and European small EVs. The trend also looks durable in parts of Asia because many consumers are highly fuel-price sensitive and Chinese-made EVs are increasingly affordable. EV sales in Vietnam, Thailand, India, and the Middle East were already on an upward trajectory before the war, with the oil shock serving to accelerate this growth rather than initiate it. The U.S. is the exception: 1Q 2026 sales were down ~27% YoY, although the decline slowed from 4Q 2025. The U.S. market could be stabilizing after the end of federal incentives, but it has not shown the same oil shock–driven reacceleration seen elsewhere (Cox Automotive, April 10).

Complications: Not just about supplying power

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It's not as simple as building a power plant and a power line to supply electricity. Labor shortages are slowing developments in general. The regulatory process is historically and necessarily extensive, just as speed-to-power is getting more important. And for AI data centers in particular, demand is substantial and rapidly increasing, but also more volatile. These constraints also make getting a physical data center site "energized" more difficult. This section explores these issues.

1. Labor constraints to remain in the foreseeable future

The labor market is tight, particularly for qualified professionals. In Uptime Institute's 2025 Global Data Center Survey, firms identified "a lack of qualified staff" as the third most important issue, with a combined ~67% of the respondents saying they are "very concerned" to "somewhat

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[5] https://www.iea.org/reports/global-ev-outlook-2026/trends-in-electric-cars

concerned", behind "cost issues" and "forecasting futures data center capacity requirements" as top management concerns.

The U.S. labor problem — a shortage of at least 222,600 workers in various trades as estimated by the Bureau of Labor Statistics (BLS) — is compounded by overlapping demand for professionals in the fields of data centers, utilities, renewables, EV charging, semiconductor manufacturing, factories, housing, transmission, and public infrastructure. Data center and power-sector employers compete for a shared set of core occupations. Remote data center development remains challenged by shortages of mechanics, electricians, plumbers, laborers, and construction workers. Attracting these trades from other industries raises development costs. (CBRE, Feb 25)

Figure 5. Labor constraints are severe in many categories involved in data center construction

Labor category	Electricians/ electrical technicians	Line worker and substation crew	HVAC and cooling specialists	Pipefitters and plumberes	Construction managers
Why needed	Medium voltage distribution, switchgear, UPS, grounding, testing	Feeders, substations, transmission/ distribution upgrades	Chillers, liquid cooling etc.	Chilled-water loops, condenser water, fuel systems, hydronics	Multi-trade planning and sequencing, safety and quality control
Projected shortage	81,000	10,700	40,100	44,000	46,800
Duration	Severe through 2030, structural through 2034	Severe through 2030	Tight through 2030	Tight through the late 2020s	Tight through 2030.

Source: Citi Research, BLS

The shortage is unlikely to clear before 2030 for structural reasons: retirements, too few apprentices relative to demand, as well as competition from grid modernization and renewable energy developments. BLS projections run through 2034 and show above-average growth for electricians, line workers, HVACR technicians, and construction managers, which implies sustained demand rather than a one-year spike. The improvement path is gradual: more apprenticeships, more prefabrication, more standardized electrical/mechanical modules, and more OEM field-service capacity, etc. But those solutions take years.

The global labor problem is uneven. Europe, the UK, Australia, and fast-growing Asian markets have serious specialist shortages (European Union, July 2, 2025; Infrastructure Australia, Nov 13, 2025). China is less constrained on construction scale, but still requires advanced operators and engineering talent.

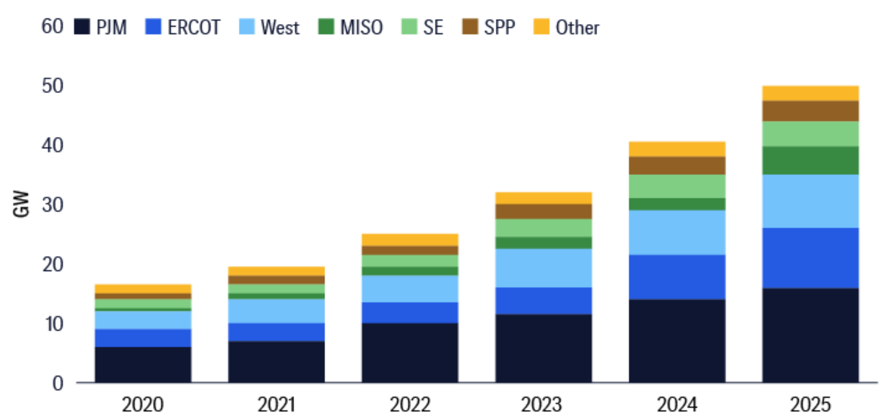
2. The complex nature of AI data center power demand

Obtaining power supply for a data center is now a complex, tall order: *Can we get firm power that is flexible enough to accommodate the massive demand and supply swings, without destabilizing the grid or triggering political backlash?* This shift is now visible in utility planning, reliability alerts, project cancellations and rate debates.

The demand and supply growth shocks go beyond just (a) volume growth; they also include (b) availability and (c) volatility issues.

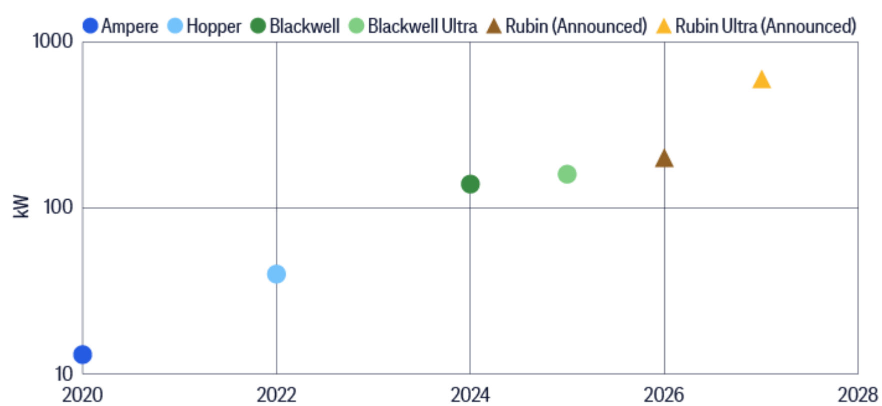
(a) **<u>Volume</u>: The U.S. Federal Energy Regulatory Commission (FERC) said more than 50GW of data center capacity was operating in the U.S. at the end of 2025, up at a ~24% compound annual growth rate (CAGR) since 2020.** The power market Midwest Independent System Operator (MISO) had the fastest growth at ~43% CAGR, followed by ERCOT (i.e., much of Texas), SPP and the Southeast. Meanwhile, capacity under construction had slipped YoY, reflecting a market increasingly constrained by permitting and equipment shortages. In addition, the power of graphic processing units (GPUs) has also surged over time as technology improves, which in turn leads to more power demand.

Figure 6. U.S. data center capacity growth has been and will continue to be strong



Source: Citi Research, FERC

Figure 7. Power of NVIDIA's AI racks has also surged over the years



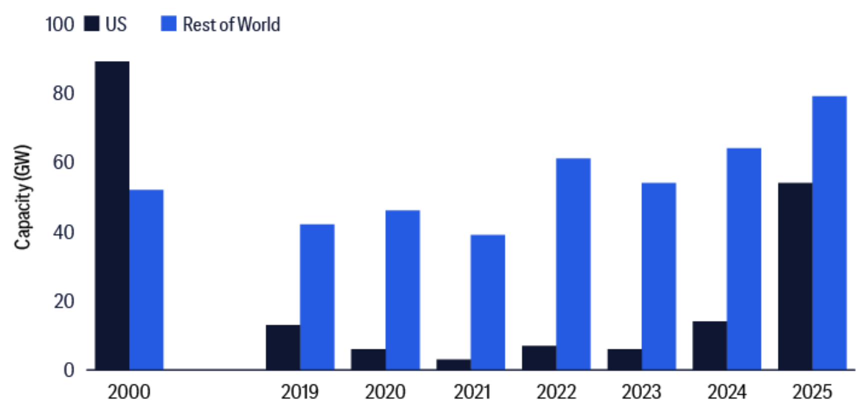
Source: Citi Research, ERCOT, IEA

(b) **<u>Availability</u>: There is a high rate of project cancellations and delays before construction.** AI demand is present, but bottlenecks prevent timely and effective energization of data centers. Cancellations jumped to 25 in 2025 from six in 2024, mainly due to power access, public pushback, and new legislation in some regions (Construction Dive, April 22). One source of power access problems is **long wait times, partly due to the approval process and equipment shortages. Long wait times have also led to duplicate requests to connect to the grid, which in turn slows approvals and physical expansions.** Switchgear, transformers, and generators may arrive so late that a site that looks viable on paper cannot energize on schedule.

At first glance, onsite generation, or bring your own generation (BYOG), has become an obvious solution, mostly using natural gas to power turbines, engines, or fuel cells.

Nonetheless, a key consideration in the current market is the durability of long-term power demand, because there was an overbuild of natural gas–fired generation around the 2000 period. The power industry remembers the early-2000s gas-build cycle, when a data center boom and power shortage fears — amid low natural gas prices — led to a wave of generation construction that later looked excessive after the dotcom bust and efficiency gains. Demand expected back then did not materialize, leaving excess capacity and ratepayer burden.

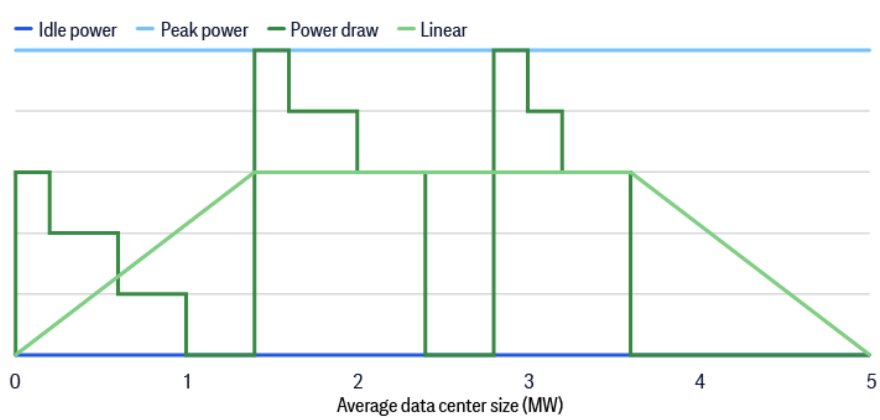
Figure 8. Natural gas turbine orders by region — can the early 2000s repeat?



Source: Citi Research, IEA

(c) **<u>Volatility</u>: The power demand swings in AI data centers come just as the penetration of intermittent renewable energy increases.** AI data centers can exhibit ~40% to 50% swings in demand over short periods depending on workload intensity, with those swings cascading into cooling and facility power requirements. Traditional utility planning methods were built around relatively stable industry and commercial load shapes. Such power demand swings also interact with weather and cooling, and vary materially across hours and seasons. Thus, developers and utilities are moving to more detailed, physics-based modeling vs. simple peak-demand assumptions.

Figure 9. Power envelope of a GPU compute cycle shows the large swings or volatility in power demand



Source: Citi Research, IEA, NVIDIA

Such operational volatility has now become a reliability issue. The North American Electric Reliability Corporation (NERC) warned of "widespread and unexpected" customer-initiated load reduction in 2024 and 2025, in which 1GW or more dropped off the bulk power system. NERC further escalated the issue into a Level 3 alert in May 2026 with required actions for certain entities. Now, the power sector needs to think about not just the quantity of demand but also its behavior — sudden drops and sharp spikes that can destabilize system operations.

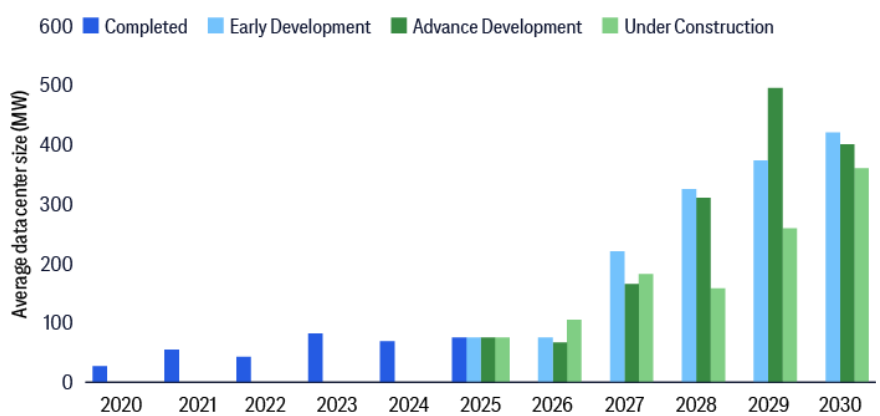
3. Regulatory and permitting process

Permitting is already a necessarily extensive process to safeguard consumers and the power network. The wait in the U.S. is worsened by some degree of duplicative applications, the need to address technical challenges for large loads, and cost-of-living concerns.

Major U.S. power markets are navigating complex regulatory and operational landscapes to accommodate growing energy demands. Citi's U.S. utilities research team examines these issues at the individual power market level later in this report.

In general, speed-to-power — against the wait for regulatory approvals, financing, and equipment procurement — has led to connection requests that are speculative, duplicated across utility queues, or dependent on BTM generation. Utilities and independent system operators have to over-study projects that may never be built. For the U.S., a study by Wood Mackenzie estimates roughly 600GW of U.S. data center projects are still searching for power, vs 183GW with construction or utility-supply agreements (Wood Mackenzie, April 28).

Figure 10. The average data center size has become much larger



Source: Citi Research, FERC

The U.S. Federal Energy Regulatory Commission (FERC) has issued a major order aimed at accelerating the connection of large power users — particularly AI-driven data centers — to U.S. electricity grids while addressing rising consumer energy costs. The plan seeks to slash approval timelines to about 90 days, compared with the years-long process that currently exists, and requires regional grid operators to update or justify how they handle large-load customers. To balance speed with grid reliability and cost concerns, data center developers may need to supply their own power during periods when the power system is under high stress, and will bear the costs of necessary grid upgrades. The move reflects surging electricity demand driven by rapid data center expansion and aims to ease bottlenecks that could slow the AI sector, while also responding to political pressure over higher utility bills and uneven grid policies across regions (Bloomberg, June 18).

At the technical level, NERC's reliability work also highlights key system-planning concerns for large loads: they can be too big and too dynamic to treat as ordinary load additions; under-forecasting them can cause insufficient generation and transmissions, and sudden loss of demand from large load can create risks in voltage and frequency that must be modeled explicitly (NERC, April 30). Grid operators and regulators are moving towards rules that define which loads need enhanced study, whether large loads must be curtailable, how they pay for network upgrades, whether they can pair with new generation, and how colocated generation interacts with wholesale transmission service.

With retail power and natural gas prices rising independently from gasoline price increases, cost of living concerns have prompted governors, consumer advocates, and public utility commissions to question whether ordinary customers should pay for substations, transmission, capacity-market costs, and gas plants built primarily for hyperscalers. Maryland's governor has pushed for reforms requiring data centers to pay for infrastructure as bills rise. Such measures can delay approvals, reshape tariffs, and force special contracts or minimum-take obligations (Reuters, May 12).

Meanwhile, the European Union's 2025 Grid Package **aims to modernize and expand its electricity infrastructure for a net-zero transition by 2050, confronting the inefficiencies of fragmented national grid planning which cost €11 billion in 2024**. The EU Grid Package faces political resistance, reflecting a tension between national sovereignty and collective efficiency, similar to challenges in U.S. grid reform. Citi's European utilities research team looks into these issues later in this report.

More broadly, satisfying strong demand and supply growth of electricity, as well as overcoming power grid constraints both raise demand for power infrastructure and generation equipment. However, very strong demand also leads to supply-chain bottlenecks that make understanding timelines and navigating these bottlenecks crucial in securing power. The following sections identify opportunities and supply-chain problems along the power value chain. There are other strategies under are also mentioned in our prior report Overcoming Gridlock: Powering Our Future (May 2024), including Dynamic Line Rating (DLR) and better forecasting and load modeling. In addition, compute can serve as flexible load: unlike other industrials, hyperscalers could shift workloads geographically and temporarily to some degree, as training or processing could shift from one data center to another, though not during a major training run. This turns computing into a somewhat flexible resource. Efficiency and cooling strategy as virtual power: Google highlights efficiency improvements, such as through lower power-usage effectiveness or PUE or compute per unit electricity consumed.

Power infrastructure equipment: Opportunities and supply-chain bottlenecks

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Opportunities in the equipment space are vast because data centers, utilities, renewable developers, gas-plant developers, and industrial projects all need similar electrical hardware. The U.S. data center electrical equipment market is projected to grow from $20 billion to $65 billion between 2025 and 2030, with critical component lead times already at 18–36 months (Wood Mackenzie, April 28). U.S. demand for generator step-up transformers has risen by more than 270% and substation transformer demand by more than 110% since 2019. Large unit lead times have reached four years and transformer prices have risen about 80% over five years (PV Magazine, May 11).

Figure 11. Power equipment demand — degree of overlap between usage by data centers and usage by renewables and storage

Equipment	Usage by Data Centers	Usage by Renewables/Storage	Degree of Overlap
Large power transformers	Campus substations, utility interconnects, step-down service	Grid substations, grid interconnects	Very large
Generator step-up transformers (GSUs)	BYOG gas power supply, fuel cells etc	Solar, wind, storage all need them	Very large
Substation transformers	Yes	Yes	Very large
High-voltage switchgear, breakers	Yes	Yes	Large
Medium-voltage switchgear	Very high use inside data centers	High use in solar, wind and storage systems	Large
HV/EHV cables and conductors	Campus and grid interconnects	Particularly for offshore wind and renewable collector systems	Large
Battery energy storage systems (BESS)	Backup or peak shaving power supply; grid services	Firming intermittent solar or wind generation	Medium-High
Gas turbines and gas engines	Yes particularly for BYOG	More at the grid level to balance intermittent renewables	Medium
HVDC converters	Indirect; useful for power-import corridors	Particularly for offshore wind and long-distance transmission	Medium
Protection relays, controls, meters	Yes	Yes	Medium
Power conversion systems/ inverters	For UPS, battery systems and some microgrid applications	Solar, wind, storage need them	Medium
Cooling equipment	Yes	No	Small
Solar modules	Only if solar is a power source	Yes	Small

Source: Citi Research

The IEA says more than 2,500GW of renewable, storage, and large-load projects are already stalled in grid queues worldwide. Meeting electricity demand through 2030 requires annual grid investment to rise about 50% from today's ~$400 billion, plus expansion of grid supply chains and workforce capacity (IEA, Feb 2026). Active U.S. power plant interconnection requests have exceeded 2,600GW, more than double the existing U.S. power-plant fleet, showing how badly renewable/storage projects already depend on grid access (Joule and US Lawrence Berkeley National Laboratory via ScienceDirect, Feb 19, 2025).

Equipment bottlenecks are particularly severe in large power transformers, generator step-up transformers, as well as in medium/high-voltage switchgear and breakers. The requirement overlap is highest where both a data center and a renewable project need to connect large blocks of power to the same grid node. A 500-MW solar+storage project, a 500-MW data center campus, and a 500-MW gas plant all need transformers, breakers, protection systems, transmission studies, substation equipment, and EPC (engineering, procurement, and construction) crew. That is why renewable developers are affected even if solar panels remain cheap.

Large power transformers are used in utility substations, data center substations, and transmission networks. There are severe shortages and limited domestic capacity, compounded by competition from utilities and generation projects. The wait time of 24–48 months is common for many large units. There are some workarounds, such as pre-buying slots, utilizing imports, refurbishing spares, standardizing designs, and using staged energization. However, these workarounds tend not to be effective because everyone is pursuing them.

Generator step-up transformers are used in natural gas-fired power plants, solar/wind/storage plants and BYOG facilities. The wait time is up to 3–4 years in some cases. Early procurement, modular generation, smaller phased plants, and reuse/refurbishment of older equipment would certainly help but would be insufficient to meet the strong demand.

Medium/high-voltage switchgear and breakers are used in data center electrical rooms, substations, and grid interconnects. Current vendor capacity constraints are exacerbated by high customization. Depending on the voltage and class, wait times can extend to 12 months, or even 30 or more. Alternatives include prefabricated electrical rooms, standardized lineups, and dual sourcing, though they are ineffective for now given the backlog.

Equity Research Views

Power infrastructure equipment prospects — a focus on transformers

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The burgeoning demand for AI data centers strains existing power grids and creates a demand surge for specialized grid equipment, from high-capacity transformers to advanced transmission infrastructure (please see our March 16 Global Electrical Equipment report for details).

1. Bright power grid equipment outlook

Global transformer supply is in shortage. According to industry consultant Wood Mackenzie, power transformers had a 30% supply deficit in 2025 (link, 14 Aug 2025). Since 2020, U.S. transformer demand has surged due to a 7% rise in electricity consumption, reversing a 1% decline between 2010 and 2020. This growth has been driven by rapid data-center expansion, a near doubling of manufacturing construction spending thanks to the tax incentives of the 2022 Inflation Reduction Act, and widespread electrification. This surge in transformer demand has created a significant supply deficit, with domestic manufacturing capacity unable to keep pace.

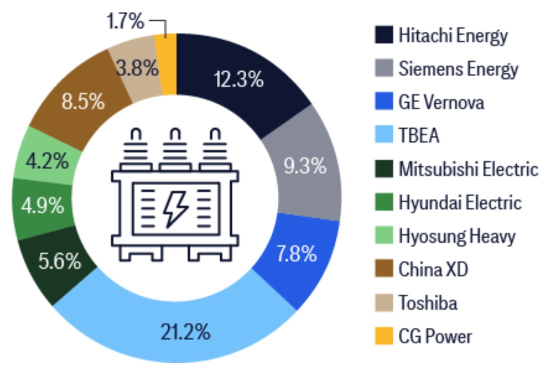
We have drawn the following conclusions based on global transformer supply and demand for global high voltage (>110kV) transformers:

Wood Mackenzie estimates a **707.5GVA annual shortfall in 2025,** based on global transformer supply of 2,358.5GVA and an assumed 30% deficit, implying that **some current demand is being deferred until additional supply comes online.** The global supply estimate is derived using data from TBEA, a leading transformer manufacturer — the largest globally and in China by capacity — with ~500GVA capacity and a 21.2% share of global capacity.

Global market shares by revenue and by capacity diverge due to significant variation in transformer specifications (ranging from 110kV to 800kV) as well as regional pricing differences, with prices in China notably lower than in overseas markets.

We forecast the annual shortage will gradually narrow from 707.5 GVA in 2025 to 143.9 GVA in 2028 and annual supply will exceed annual demand from 2029.

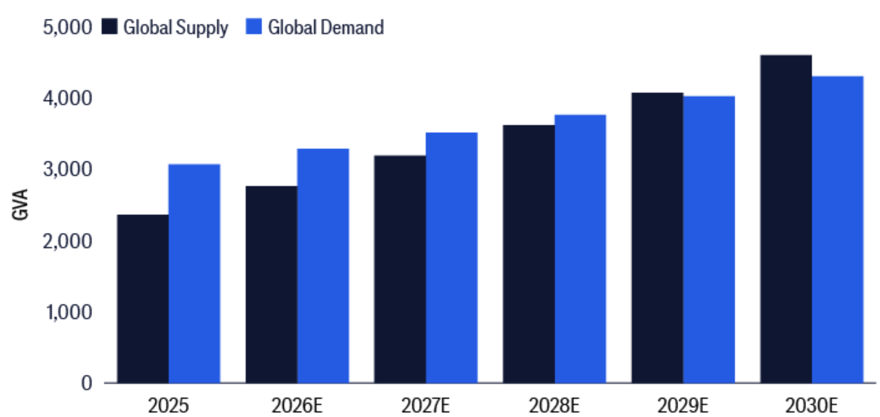
Figure 12. 2025 market share by capacity total 2,358.5GVA



Source: Citi Research, Company Reports

However, we estimate the cumulative shortage will keep rising and peak in 2028 at 1,698.8 GVA — equal to 47% of annual supply in that year — and persist in the run-up to 2030. For supply growth, the top-10 companies were dominant, representing 79.4% of global market share in 2025 and a projected 72.6% in 2030 in terms of high-voltage transformer production capacity. We have assumed demand growth to rise to a 7% CAGR in 2025–2030, largely in line with the forecasts of industry consultant Arizton Advisory & Intelligence.

Figure 13. Global supply vs. demand



Source: Citi Research, Company Reports

U.S. utilities are increasingly turning to the import market to meet project timelines. According to industry consultant Wood Mackenzie, imports account for an estimated 80% of U.S. power transformer supply and 50% of distribution transformer supply. This growing dependence on foreign-sourced equipment highlights the scale and speed of demand growth across the industry. This trend is most evident in power transformers and generation step-up transformers, where U.S. demand is estimated to have increased 116% and 274%, respectively, since 2019.

Wood Mackenzie estimates that power transformers are in a supply deficit of 30% based on annual supply and demand estimates in 2025. The market imbalance has increased competition for available production slots, leading to a large rise in lead times and prices in most transformer categories. Since 2019, unit costs have risen by 45% for generation step-up transformers, 77% for power transformers and 78–95% for distribution transformers depending on specs, though higher-voltage units have seen larger U.S. dollar-based cost escalations due to heftier baseline costs.

These recent market trends have underscored the deep supply-chain challenges facing the transformer sector stipulated by the industry consultant, from raw-material shortages and technical labor limitations to the evolving U.S. policy landscape. The U.S. relies on a single domestic supplier of GOES (grain-oriented electrical steel), AK Steel, forcing many OEMs to source raw GOES or semi-finished transformer cores abroad.

Wood Mackenzie notes that another issue is the specialist technical labor required to manufacture transformers. OEMs have cited labor shortages as a key reason for not scaling up output to meet the growth in demand. Copper windings, meanwhile, have emerged as another potential bottleneck to transformer production, due to their highly technical manufacturing process and niche specifications, and the U.S. government's 50% copper tariffs are likely to worsen the issue.

Transformer costs are likely to continue to increase with the imposition of U.S. trade tariffs, said the industry consultant, as the industry is highly dependent on imports for both finished equipment and underlying components and commodities. Not only will the average cost per transformer increase significantly due to the tariffs, but lead times also will escalate as imports from certain high-tariff countries become cost prohibitive. Many of the largest transformer OEMs have announced capacity expansions since 2023 to address the shortage in the North American market. But Wood Mackenzie says the rate of demand growth will require even more investment to rebalance the market.

2. Global annual high voltage transformer supply below annual demand in the run-up to 2029

We forecast global high voltage power transformer supply to rise 59% in 2025–2028 and 95% in 2025–2030, taking into account the expansion plans of the global top-10 suppliers (see Figure 14). We cover the top-3 in further detail below.

Figure 14. Global high voltage (>100kV) power transformer supply (based on capacity) and demand (Unit: GVA)

Ranking	Company	Country	2025 market share in terms of revenue	2025 market share in terms of supply	2025	2026E	2027E	2028E	2029E	2030E	2025-28E	2025-30E
1	Hitachi Energy	Japan	16.50%	12.30%	290	330	380	450	480	500	55%	72%
2	Siemens Energy	Germany	14.20%	9.30%	220	240	270	305	370	440	39%	100%
3	GE Vernova	The US	11.80%	7.80%	185	200	220	240	250	260	30%	41%
4	TBEA	China	10.10%	21.20%	500	500	620	620	700	750	24%	50%
5	Mitsubishi Electric	Japan	7.30%	5.60%	132.1	149.3	166.5	172.3	198.1	227.8	30%	73%
6	Hyundai Electric	Korea	6.50%	4.90%	115.6	121.3	150.2	165.3	190	209.1	43%	81%
7	Hyosung Heavy	Korea	6%	4.20%	100	120	144	185	212.8	234	85%	134%
8	China XD	China	5.20%	8.50%	200	350	350	380	380	450	90%	125%
9	Toshiba	Japan	4.70%	3.80%	89.6	100	115	134.4	150	161.3	50%	80%
10	CG Power	India	3.90%	1.70%	40	40	40	85	85	100	113%	150%
Rest of World					486.2	607.8	729.3	875.2	1050.2	1260.3	80%	159%
Global Supply					2358.5	2758.4	3185	3612.2	4066.1	4592.5	53%	95%
y/y						17%	15%	13%	13%	13%		
Top 10 market share			86.20%	79.40%	79.4%	78.0%	77.1%	75.8%	74.2%	72.6%		
Global demand					3066	3280	3510.3	3756	4019	4300.3		
y/y						7%	7%	7%	7%	7%		
Annual shortage					-707.5	-522.2	-325.2	-143.9	47.2	292.3		
Cumulative shortage					-707.5	-1229.7	-1554.9	-1698.8	-1651.6	-1359.3		
vs. annual supply					30%	45%	49%	47%	41%	30%		

Source: Citi Research, Company Reports, Wood Mackenzie, Arizton Advisory & Intelligence

1. **Hitachi Energy** expects its capacity expansion for high voltage transformation production to peak in CY2027–2028. It will spend US$1.655 billion to build a new high voltage transformer production plant to be completed in 2025–2027, with the capex allocated as follows: 40–42% in the U.S., 33–36% in Europe and 22–27% in Asia. We assume a cost of US$450-500 million to build a new GVA production line for making high voltage transformers of around 40-50 GVA per annum in full utilization. We assume the new capacity to be added gradually over four years from 2025 to 2028, taking into account both the capacity addition schedule and the time taken for training skilled labor. The current plan targets a doubling of sales by CY2030.[6]

2. **Siemens Energy** will add new transformer production capacity primarily in Germany, the U.S. and India. For Germany, it will invest €220 million (about US$260 million) in a new GVA transformer production line to be completed in 2028. In the U.S., it has invested US$1 billion in two new GVA transformer production lines to be completed in February 2026. In India, it will invest a total of US$291 million to add transformer production capacity mostly in 2030–2032. Taking all these into account, the company is increasing transformer capacity by 85 GVA to 2028, which translates into a 30-50% capacity increase. The expansions would also skew more of the capacity expansion to 2027 vs. 2026 given the timing of the opening of the transformer factory in Charlotte, North Carolina (likely starting production in late 2026). And it aims to double its capacity by 2030.[7]

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[6] Based on company press releases: 30 Sep 2025, 18 Nov 2025, 25 Apr 2025, 10 Mar 2025

[7] Based on company and industry sources: ModernPower Systems (9 Sept 2025), T&D India (14 Feb 2026), Mercom (19 Feb 2026), Siemens Energy (3 Feb 2026)

Figure 15. Capex to build new high voltage transformer production capacity (>220kV) per 1GVA

Region	Capex* (US$m)	Vs. the capex in the US
The US	450-500	N/A
Japan	350-400	15-20% lower
Korea	300-350	25-35% lower
Europe	320-380	20-30% lower
China	250-300	35-45% lower
Middle East	380-430	10-15% lower

Source: Citi Research

3. **GE Vernova** (GEV) is in the process of adding significant transformer capacity. It has publicly stated that it is doubling its output of transformers and switchgear globally from 2024 through 2028 with most of that coming from more shifts and more investments into existing operations. GEV announced adding significant capacity in Stafford, UK, to support fast-growing High-Voltage Direct Current (HVDC) transmissions systems while adding to its capacity in other global locations, including a significant investment plan to expand switchgear and transformer manufacturing capacity in India. GEV's recently closed acquisition of the remaining 50% stake in Prolec GE that it did not own should also add significant transformer capacity, with GEV management recently stating that it would incorporate an additional $1 billion of incremental capex through 2028 from Prolec GE to support increased transformer production.

Each GVA line could have capacity between 65-100GVA for greenfield development. For expansions of existing plants, the capex could be just 30-50% that of newly built plants for the same capacity addition.

We calculate the capex needed to build new high voltage transformer production capacity (>220kV) at US$450-500 million per 1 GVA of capacity in the U.S., based on our due diligence with various transformer makers. And the capex amounts are lower than the U.S. by 15-20% in Japan, 25-35% in Korea, 20-30% in Europe, 35-45% in China and 10-15% in the Middle East. The capex per 1 GVA capacity needs to increase by 20-30% for plants used to make ultra-high voltage (800kV+) transformers.

3. Rising global transformer demand

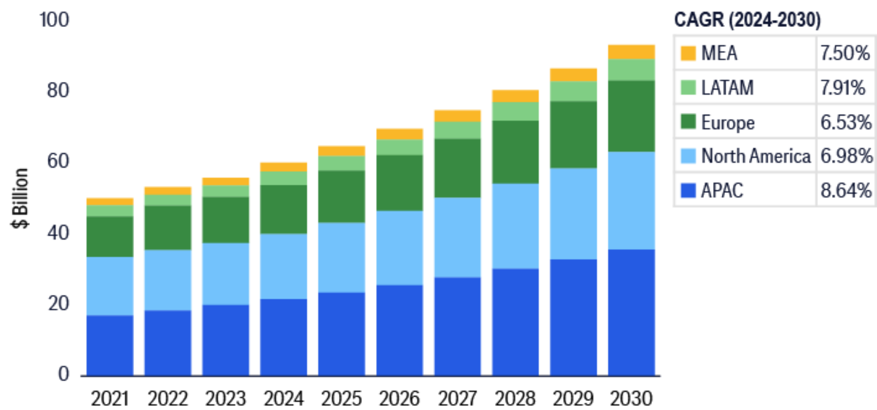
The global transformer market is estimated to rise at a 7.6% CAGR from US$60.2 billion in 2024 to US$93.5 billion in 2030, according to industry research firm Ariston Advisory. The global transformer market growth is primarily boosted by more demand for power transformers resulting from the following:

- **Rising energy share from renewables globally**. Ariston Advisory forecasts around 4,000GW of new power generating capacity in 2025–2027 driven by more renewable energy uses and catering to increasing electricity demand.

- **Replacement of aging infrastructures** in North America. The average age of transformers used in the U.S. is 38 years, exceeding their designed useful lives of roughly 30–35 years.

- **Sizeable development in such regions** as the Middle East, Southeast Asia, Latin America and Africa, with the majority of their power transformers imported (e.g., 90% for the Middle East).

Figure 16. Transformer Market by Region (US$bn)



Source: Citi Research, Arizton Advisory

The worldwide transformer market includes the design, development, production and after-sales service of transformers for electricity transmission and distribution. The market therefore comprises various products ranging from power transformers, distribution transformers, instrument transformers as well as specialty transformers for such specific applications as renewable energy integration, industrial automation and rail electrification. And this market expansion is a consequence of more demand for energy usage, increasing use of renewable energy, and further advancement in grid modernization.

The following market trends are causing more transformer demand:

- **More digitalization and microelectronics**, in particular AI, have led to more electrification and hence more transformer demand. Based on IEA estimates in April 2025, electricity consumption of data centers globally would exceed 1,000TWh in 2026, representing 6% of energy consumption worldwide. Also, according to IEA, U.S. power consumption by data centers will represent half of the national incremental electricity demand in the run-up to 2030.

- **More decentralized energy generation** driven by more usage of smart grid and energy storage technologies as well as distributed renewable energy projects, which need power-efficient and smaller transformers, and distribution transformers adjacent to wind, solar, and EV charging stations.

- **Accelerated global electricity demand growth.** IEA projects 85% incremental global electricity demand in 2024–2027 from emerging markets with the biggest contributions from China, India and Southeast Asia. These regions are making significant investments in electricity infrastructure to ensure reliable power supply for expanding digital hubs, particularly as grid pressures intensify.

In 2024, global power transformer production was 1,936k MVA, while global transformer production capacity was 3,200k+ MVA. Arizton Advisory estimates that actual production was less than capacity owing to:

- **Supply-chain bottlenecks:** Shortages of key materials such as electrical steel and insulation materials have slowed manufacturing schedules, despite available plant capacity.

- **Long lead times and backlogs:** Many factories are operating at high utilization, but with long order backlogs, hence taking multiple years for delivery. This creates a gap between installed capacity and annual output.

- **Project-specific customization:** Power transformers are highly engineered and customized. Testing and validation cycles extend production times, preventing full annual capacity utilization.

- **Labor and workforce constraints:** Skilled workforce shortages, particularly in winding and assembly, continue to constrain throughput in several regions.

- **Regulatory and project delays:** Even when transformers are produced, delays in grid projects or approvals can stagger shipments and recorded production.

Transformer delivery cycles have stretched from months to multiple years, supporting the view that annual production volumes will remain below production capacity.

Wood Mackenzie estimates that, in 2025, power transformers and distribution transformers had deficits of 30% and 10% respectively in the U.S. (link). The U.S. needed around 25,000 units of power transformers in 2025; in other words, the shortage was 7,500 units last year.

Amid this shortage, power transformers and generation step-up have maintained average lead times (the period from receiving new orders to delivery) consistently above 100 weeks (or roughly two years) since 2023. The institute expects that the shortfall will persist in the run-up to 2030 despite many of the largest transformer makers announcing local production capacity additions, including Siemens AG's US$150 million investment in a new power transformer facility in Charlotte, North Carolina. Much incremental transformer supply in the U.S. is from imports — in 2025, imports are estimated to have accounted for about 80% of U.S. power transformer supply and 50% of the distribution transformer supply. Chinese-built transformers are not commonly used in the U.S. owing to cybersecurity concerns. On the back of this trend, we expect HSH to export more transformers made in Korea to the U.S.

Figure 17. Production capacity of key global transformer makers (MVA/units)

Company	Est. annual production capacity	Key manufacturing location	Recent developments
Hitachi Energy	400,000+	Global	On 4 Sep 2025, > US $ 1bn spending to expand production in the US was announced
TBEA	420,000	China	Expanded production capacity to 420,000 in 2024
Siemens Energy	250,000	Germany, the US, India, Croatia	On 5 Sep 2025, US$235m capex to lift transformer production capacity in Nuremberg, Germany by 50% was announced
GE Vernova	170,000+	The US, India, Brazil	In Jan 2025, a US$600m investment plan in US factories & facilitates in 2025-26E were announced
Mitsubishi Electric	100,000-150,000	Japan	In Oct 2024, a US$80m investment in a switchgear factory in Western Pennsylvania, US, was made
Hyosung Heavy	80,000-120,000	South Korea	Plans to nearly double its US output to > 250 units pa by 2027 E
CG Power & Industrial	40,000+	India	A US $80m investment was announced in 2023 to add transformer annual production capacity by 45,000 MVA to 85,000 MVA
WEG S.A.	50,000-85,000+	The US, Brazil, Mexico, Colombia	
Schneider Electric	50,000-70,000+	Global	In Feb 2025, an investment plan to add medium power transformer manufacturing capacity in India was announced
Toshiba	42,000	India	In Apr 2025, an US $60m investment was announced to expand manufacturing capacity at its Rudraram plant

Source: Citi Research, Arizton Advisory

Power generation equipment: Opportunities and supply-chain bottlenecks

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Commodities Strategy Views

Speed-to-power concerns drive demand for power generation equipment — from more efficient combined cycle gas turbines, to smaller, less efficient reciprocating engines, aeroderivative turbines and fuel cells, to the use of renewables and energy storage systems (ESS). This section looks at strategies by hyperscalers and focuses on various BYOG options.

1. Strategies adopted by hyperscalers

Hyperscalers (AWS, Microsoft, Google, Meta) have converged on a "stack" of strategies to secure speed-to-power at scale. For now, they move to where power is, finance power, or bring power, also known as "bring your own generation," or BYOG. Hyperscalers cannot simply move more AI abroad because constraints outside the U.S. include grid access, sovereign-control requirements, chip-policy risk, and local regulations.

- **Colocation with generation to bypass local grid bottlenecks:** Colocation, especially nuclear or existing thermal power generation, has been attractive because it helps minimize exposure to local constraints, but it raises questions about transmission cost allocation. It is a "has been" because most sites have already been identified. Examples include Microsoft-Constellation's 20-year PPA supporting the restart of Three Mile Island Unit 1 (835-MW); Meta-Constellation on a long-term deal associated with the Clinton Clean Energy Center (AP, Nov 19, 2025; Guardian, Jun 4, 2025).

- **Underwriting of supply such as PPAs or direct investments:** Hyperscalers still use large-scale power purchase agreements (PPAs) for wind/solar to claim clean energy attributes and catalyze new builds. Amazon has described its global PPA approach as central to meeting its renewable goals (Amazon, July 2024; Amazon, Feb 2026). Some hyperscalers also pair PPAs with 24/7 carbon-free energy and "hourly matching" contracts (newer, more demanding procurement) to make them deliverable. This procurement style helps drive investments in firming (storage, geothermal, nuclear, hydrogen-ready assets), as hourly matching is harder than annual matching. Hyperscalers also pursue venture investing, such as in long duration energy storage (LDES), geothermal, or advanced nuclear pathways.

- **Behind-the-meter/onsite generation and storage (BYOG):** Hyperscalers or their colocation partners have become more focused on onsite generation and storage because of the multi-year wait for power access or infrastructure upgrades. While it improves speed-to-power and reliability *locally*, many of these options increase the dependence on natural gas infrastructure.

2. BYOG options

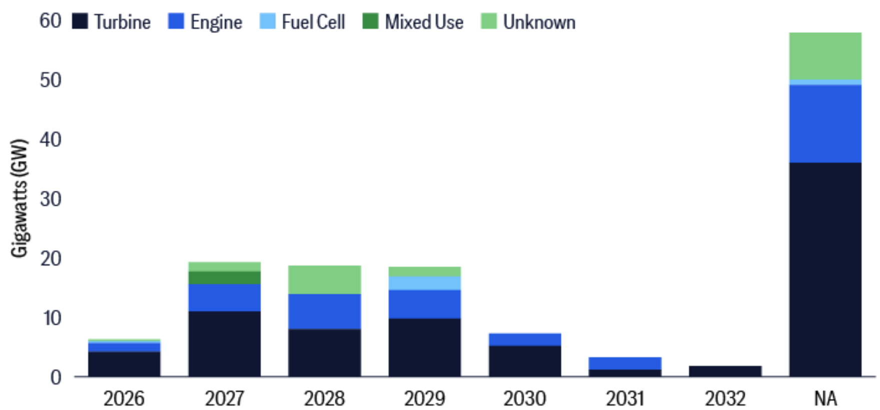
Longstanding key advantages of the current centralized bulk power system (BPS) with a large transmission network are supposed to be its cost and space efficiency: large power plants generally have lower

generation costs, but are too large to be in population centers, where power demand is large.

However, if the speed-to-power problem does not improve significantly, other power consumers could increasingly adopt BYOG as well. Data centers are large loads that can afford to have their own generation onsite. The move to distributed generation is possible because renewables (mostly solar) + battery systems are small enough that households or communities can have their own generation.

There are multiple options, including (a) gas turbines, (b) reciprocating engines, (c) solid oxide fuel cells (SOFC), (d) diesel backup generators, (e) energy storage systems, and (f) other longer-term technologies. Many options use natural gas as an energy source. Our prior report [GPS: Natural Gas (2021)] discussed how gas can find its product-market fit (PMF).

Figure 18. Announced onsite gas power capacity by technology and timeline



Source: Citi Research, BNEF

(a) Natural gas turbines: There are larger, combined-cycle gas turbines (CCGT) or simple-cycle gas turbines for behind-the-meter usage. Each of these turbines is hundreds of megawatts (MW) in size. They have the lowest $/MWh for long-run power generation. They are more attractive where a data center cluster can justify a full plant.

Aeroderivative gas turbines (e.g. LM-class), such as those by GE Vernova, are packaged turbines used for faster deployment and peaking, backup or even prime power generation. They are often 25 to 100-MW in size. They have high power density and are good for campuses with limited footprint. They can be configured for grid-parallel operation.

(b) Natural gas reciprocating engines: Recip engines, as they are also known, are multiple containerized gas engines, often in 5 to 20-MW blocks that can be aggregated into gas plants that are 50 to more than 300 MW in size. They are fast to procure and install compared to large turbines and long transmission upgrades. They are modular in that a developer can add blocks as the campus grows. They have good part-load efficiency and can ramp up quickly.

Pairing this system with other energy storage or generation is common. Pairing with batteries, such as lithium-ion or UPS (uninterruptible power supply) that are mostly lead-acid, allows the system to ramp up and down much faster. This pairing can help smooth generation and match demand, particularly when data center power demand can have very large swings. The combination exists such that recip engines and other gas turbines or solar are installed at the same campus.

A subset is four-stroke engines used for data center power, which are usually medium-speed reciprocating gas or dual-fuel engines derived from marine, island-grid, or industrial power applications. They are not typically ship engines removed from vessels but are stationary power versions of engine platforms from suppliers. Compared with conventional diesel backup generators, they are better for longer-duration or even continuous operation, especially fueled by natural gas.

(c) Solid oxide fuel cells (SOFC): SOFCs are modular onsite fuel cell blocks commonly fueled by natural gas. They are used for prime power or grid-parallel supply. Bloom Energy is the most visible supplier in U.S. data centers. With their modular construction, SOFCs can scale in increments. They can have high uptime and a cleaner local profile vs. diesel. Yet SOFCs still depend on the availability of gas, and the economics depend on utilization, spark spread and contract structure.

(d) Diesel backup generator: These are the traditional standby diesel generators, often multiple 1-3 MW units per building or block, paired with UPS. These diesel generators have the highest certainty for emergency backup reliability, as well as a mature supply chain and servicing. However, there is community pushback, including noise quality and air quality, such that they often have tighter air permitting (Inside Climate News, Nov 2025),

(e) Battery energy storage + UPS with or without renewables: Given the rapid swings for some AI data center power demand and the intermittency of renewables, batteries can help to smooth power delivery. However, rapid cycling reduces the lifespan of batteries. Batteries can also provide short-term power as other engines or turbines ramp up.

As a standalone application, external battery packs or battery energy storage systems (BESS) can extend runtime to several hours. A BESS generally uses lithium-ion batteries and can provide 1–4 hours of power. These systems are used for ramp smoothing, black start, peak shaving, or demand change management. UPS batteries, often lead-acid, can ramp up essentially immediately and provide power for minutes.

(f) Longer-term BYOG options: Nuclear, including small modular reactors (SMRs), is often discussed by the technology community. But it is not an option for quick deployment. Large nuclear reactors take years to build, at times 10 years or more. SMRs still need time to commercialize, since some are only under construction in western markets.

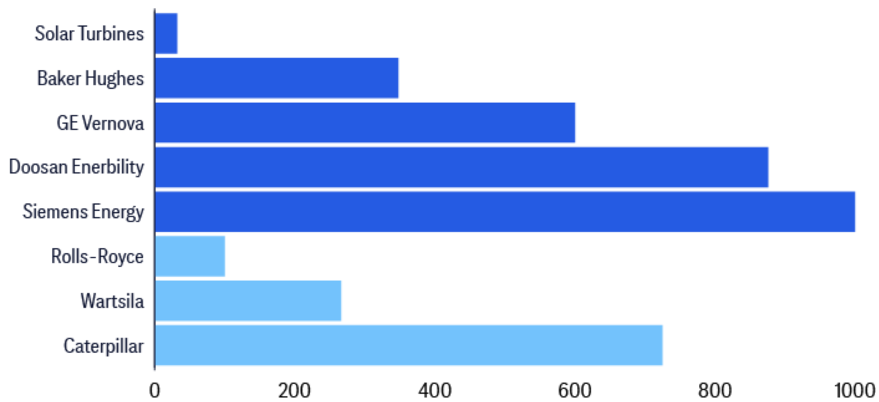
Common deployment strategies in the U.S. currently include: natural gas recip engines + BESS or SOFC + BESS, which can very quickly deliver modular power supply; aeroderivative gas turbines + BESS, which also optimizes for space to deliver high power density; diesel generators, which likely see reduced runtime due to air emissions; and solar + BESS pairing.

Equipment manufacturers accelerating investments to deal with the supply-chain bottleneck

Power equipment manufacturers are facing significant challenges in scaling production of turbines and engines to meet rising order volumes. Lead times for some equipment are long. BNEF estimates the gas equipment lead times to be between 12–29 months for engines, 12–18 months for light industrial turbines, 15–36 months for aeroderivative turbines, and 36–84 months, or post-2030, for heavy-duty turbines.

The data center boom is driving a massive surge in demand for onsite gas power, with 128 global projects totaling 133GW announced. Gas turbines dominate at 77GW, followed by engines at 33.8GW, with GE Vernova, Doosan Enerbility, and Siemens Energy among key suppliers linked to 49GW. This demand boosts manufacturers' revenues, with GE Vernova projecting data centers will comprise 25% of its orders by 2026, and others including Rolls-Royce and Caterpillar reporting substantial YoY growth and order intake increases from this sector.

Figure 19. Capital allocated for gas equipment-related manufacturing expansion



Source: Citi Research, BNEF

Gas power equipment manufacturers are committing nearly $4 billion, led by Siemens Energy ($1 billion), Doosan Enerbility, and Caterpillar, to significantly boost production, primarily for data center demand. Turbine OEMs are investing more heavily, with some, including Caterpillar, building new factories, while others, such as GE Vernova and Baker Hughes, expand existing ones. Manufacturers plan output increases of 20–200%, largely targeting the U.S. market, with examples including Caterpillar doubling engine capacity by 2030 and Baker Hughes doubling NovaLT turbine production by 2027.

Figure 20. Selected planned expansions of gas equipment manufacturing

Company	Capital ($ million)	Details	Target year
Baker Hughes	348	Double manufacturing capacity by 1H 2027 for NovaLTTM turbines	2027
GE Vernova	600	Raise capacity to 20-GW by mid 2026 and 24-GW by 2028 for gas turbines, with expansions for switchgear etc as well	2025-2028
Mitsubishi Heavy Industries	NA	Double within two years gas turbine capacity (planned)	2027
Siemens Energy	1000	Boost capacity from 17-GW in 2024 to 30-GW by 2030 for turbines (planned)	2030
Caterpillar	725	Double capacity by 2030 for engine production	2024-2030

Source: Citi Research, BNEF

Equity Research Views

Andrew Kaplowitz
Vladimir Bystricky
U.S. Industrials and Conglomerates Equity Research

A focus on gas turbines, engines and other on-site generation

Given rapid expected growth in global electricity demand in the coming years, demand for power generation equipment appears to be in the midst of a multi-year growth cycle, with industry participants seeking to ramp availability of power generation equipment across a broad range of equipment types. Strong demand signals and a combination of both near-term demand growth and expected long-term power generation needs we think are driving a relatively broad supply-side response, with industry participants seeking to address growing electricity needs through a multi-pronged approach including "shorter-cycle" (more rapidly deployable) power generation equipment and "longer-cycle" power generation equipment such as large heavy duty gas turbines.

Global gas turbine orders, which we view as a reasonable proxy for global power equipment demand (albeit exclusive of non-gas fired equipment such as coal, renewables, and nuclear), after averaging ~42GW per year in the five-year period from 2019–2023, increased to ~58GW in 2024, followed by a spike to ~100GW in 2025, and remain elevated into 2026 with 1Q global gas turbine orders of ~29GW, highlighting elevated strength in global demand for power generation equipment. Rapidly growing demand for gas turbines appears to be outpacing near-term supply capacity and resulting in extended delivery timelines or multi-year backlogs for OEMs; GEV for instance indicated in April 2026 that lead times for large gas turbines at the time were ~three years.

At the same time, data center owner/operators seem focused on scaling their data center infrastructure investments, and focused in part on "speed to market" — operationalizing their investments to support revenue generation. As such, we have observed increasing demand for "shorter-cycle" power equipment solutions that can potentially be more rapidly procured and deployed "behind the meter" to supply near-term data center related power demands, with companies such as Caterpillar, Cummins, Siemens, and others having all reported increased demand and capacity expansions under way for their power solution offerings for data centers. Notably, these "shorter-cycle" offerings span a range of technologies including diesel reciprocating engines, gas reciprocating engines, and smaller (vs. utility scale) gas turbines. Our sense is that market demand for these "shorter-cycle" power generation equipment offerings could remain relatively robust over the coming years, led by "speed to market" imperatives amongst data center owners in the near-term that remain supportive of demand for power generation assets that are relatively quickly deployable and supportive of "behind the meter" approaches to powering data centers.

That said, we do note that "shorter-cycle" power generation equipment, though in many cases relatively flexible and fast to deploy, does involve some tradeoffs in comparison to larger-scale heavy-duty gas turbines that may offer lower operating costs and generate lower carbon emissions.

Additionally, while a significant portion of near-term demand for power generation equipment is associated with data center related electricity needs, broader global trends noted elsewhere in this report such as electrification and energy security imperatives are also contributing to demand for power generation equipment.

As such, “longer-cycle” heavy duty gas turbines also remains poised to reflect elevated demand in the coming years, with OEMs in this segment of the market also adding capacity to meet growing demand.

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Caterpillar (CAT) and Cummins (CMI) are two of the largest and most important suppliers of onsite power generation equipment to the data center market, and both companies are aggressively expanding capacity to meet surging demand (read more here). CAT's product portfolio spans gas turbines and large natural gas and diesel reciprocating engines, giving it exposure to both the prime power (onsite, behind-the-meter) and backup power segments of the data center market. CMI, by contrast, currently sells exclusively large diesel reciprocating engines into the data center backup power market, though the company is developing a natural gas reciprocating engine product targeting the prime power market, expected to be available in 2028.

Both companies have announced significant capacity expansion plans: CAT is targeting ~65 GW of combined turbine and large engine capacity by 2030, up from ~22 GW in 2022, while CMI is targeting ~55 GW of large engine capacity by 2030, up from ~26 GW in 2022. We estimate that approximately 40 GW of CAT's 2030 capacity and approximately 30 GW of CMI's 2030 capacity will be directed toward the data center market. Translating this capacity into revenue, we forecast CAT's data center revenues to reach approximately $20bn by 2030, which we think would imply roughly a 4x increase in its power gen revenues from 2024 levels, compared to its >3x target, while we forecast CMI’s data center revenues to reach approximately $12.5 billion by 2030, compared to its $9 billion+ target.

With a significant capacity buildout expected across the industry through 2030+, we have observed a growing debate around whether there will be enough incremental demand to satisfy that supply. Overall, we see little risk of industry oversupply through 2030 for both onsite backup and prime power. Using what we deem as our most realistic assumptions for what industry capacity utilization could look like, we believe the industry supply/demand dynamics could start to loosen in the early 2030s. Please see the report US Machinery - CAT & CMI Data Center Note 2: Digging into Supply/Demand (June 24) for details.

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U.S. Oil and Gas and Oil Field Services
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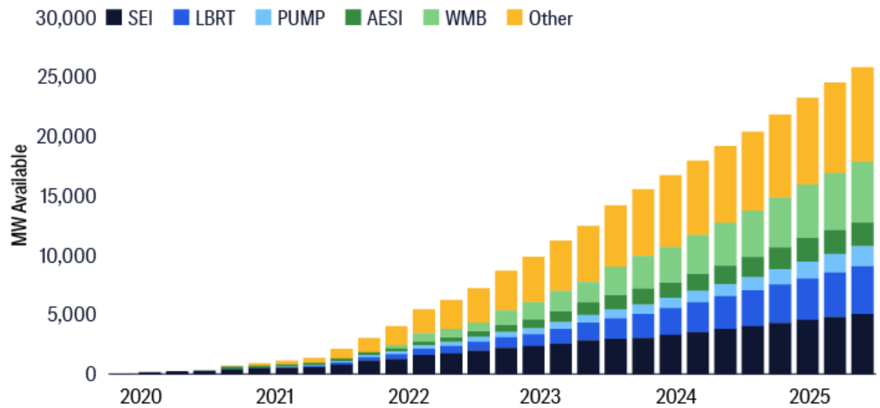
Separately, we believe the market for behind-the-meter power from third-party suppliers will grow toward 25GW in 2030, with the vast majority of capacity deployed at data centers. With uncertainty toward timing of grid connect and rising residential utility bills, data center operators are seeking power solutions from an unlikely source — the oilfield services and midstream industries.

Figure 21. Behind-the-meter power capacity outlook



Source: Citi Research

Over the past decade, oilfield services companies developed and deployed small-scale mobile turbines as well as gas reciprocating engines to convert cheap, abundant natural gas into power to run hydraulic frac pumps. The midstream industry also has experience running gas turbines at compressor stations as they mobilize gas throughout the U.S. Now, oilfield services and midstream companies are looking to deploy turbines and gas recip engines at data centers for primary power. Initially viewed as bridge power, contract terms have improved with recent deals extending for 10 years and include exit penalties for customers. Power generation providers are also expanding their offerings to include ancillary equipment to run the microgrids, handle fluctuations in load and mitigate emissions. At times, they are also in charge of supplying natural gas.

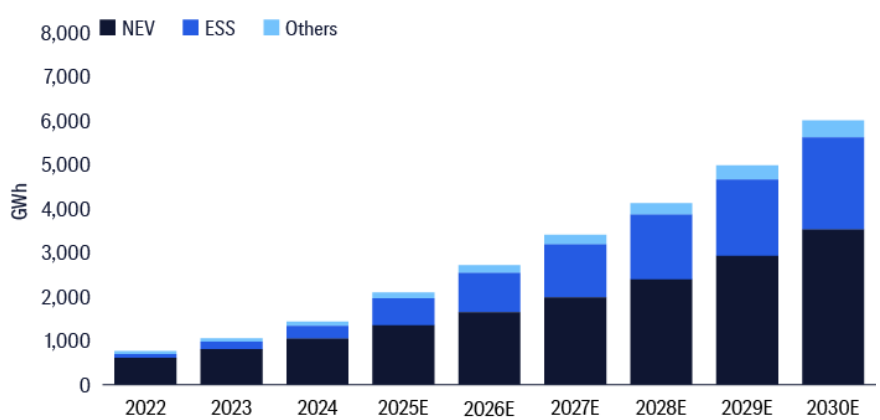
A focus on batteries and energy storage systems (ESS)

Jack Shang
Cynthia D Wu

Asia Metals and Mining Equity Research

We see growing likelihood that governments globally will accelerate the shift toward renewables and electrification to strengthen energy security, particularly in light of the Iran conflict. The strategic importance of the Middle East underscores the risks of continued dependence on imported oil for remaining net importers. Alongside net-zero goals, this is creating sustained tailwinds for higher NEV penetration — particularly outside China — and stronger global demand for Energy Storage Systems (ESS) over the long term.

Figure 22. Global battery demand forecast (2025-2030)



Source: Citi Research

We project global battery demand to reach 6,005 GWh by 2030, representing a 23% CAGR over 2025–2030, with NEV battery demand growing at a forecast 21% CAGR to 3,523 GWh and ESS battery demand expanding at a faster forecast 28% CAGR to 2,093 GWh over the same period; as a result, the importance of ESS is expected to rise, accounting for around 35% of total demand by 2030, supported by its role in peak shaving, energy arbitrage, and enhancing energy security.

NEVs have undergone decades of technological advancement, with continued improvements in versatility, efficiency, and cost competitiveness driving their adoption across global markets. Looking ahead, leading battery manufacturers are expected to further reduce costs through higher yield rates, better asset utilization, and increased automation, while also enhancing performance — such as extending driving range, enabling faster charging, and improving operation in low-temperature conditions.

Figure 23. Citi long-run demand assumptions for EV sales and penetration ratio by region

EV sales (k units)	2021	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E	2025-28E	2025-30E
China (incl. exports)											55%	72%
BEV	2,734	5,015	6,142	7,115	9,449	10,921	12,567	14,880	17,013	19,580	39%	100%
PHEV	576	1,486	2,731	5,119	5,857	5,835	6,594	7,137	7,644	7,614	30%	41%
Total EV	3,310	6,502	8,873	12,234	15,306	16,756	19,161	22,017	24,656	27,194	24%	50%
YoY	182%	96%	36%	38%	25%	9%	14%	15%	12%	10%	30%	73%
EV penetration rate	16%	28%	35%	45%	52%	58%	64%	72%	79%	85%	43%	81%
US											85%	134%
BEV	455	728	1,130	1,222	1,233	1,312	1,422	1,449	1,652	1,863	90%	125%
PHEV	175	185	295	315	261	295	313	336	429	437	50%	80%
Total EV	630	913	1,425	1,537	1,494	1,606	1,735	1,784	2,081	2,301	113%	150%
YoY	106%	45%	56%	8%	-3%	8%	8%	3%	17%	11%	80%	159%
EV penetration rate	4%	7%	9%	10%	9%	10%	10%	11%	12%	13%	53%	95%
Europe												
BEV	1,111	1,436	1,800	1,794	2,090	2,293	2,740	3,243	3,844	4,473		
PHEV	1,028	1,000	974	899	1,000	1,182	1,391	1,506	1,595	1,700		
Total EV	2,139	2,436	2,773	2,693	3,089	3,475	4,130	4,749	5,438	6,174		
YoY	61%	14%	14%	-3%	15%	12%	19%	15%	15%	14%		
EV penetration rate	18%	22%	22%	22%	26%	30%	36%	41%	48%	54%		
RoW												
BEV	309	388	376	579	579	834	1,160	1,534	2,044	2,437	90%	125%
PHEV	58	77	81	93	127	168	201	283	348	413	50%	80%
Total EV	367	465	457	672	706	1,002	1,360	1,817	2,393	2,850	113%	150%
YoY	71%	27%	-2%	47%	5%	42%	36%	34%	32%	19%	90%	125%
EV penetration rate	1%	2%	2%	3%	3%	4%	6%	7%	10%	11%	50%	80%
Global EV	6,445	10,315	13,529	17,136	20,596	22,839	26,387	30,367	34,569	38,518	90%	125%
YoY	113%	60%	31%	27%	20%	11%	16%	15%	14%	11%	50%	80%
Global PV sales	72,480	75,592	79,170	80,483	82,489	81,250	82,357	83,551	84,904	86,514	113%	150%
EV penetration rate	9%	14%	17%	21%	25%	28%	32%	36%	41%	45%		

Source: Citi Research, ICCSINO, Real Lithium, Company Reports

We expect global NEV penetration to rise further from 25% in 2025 to 45% by 2030. Within this, Europe is projected to see the fastest growth, reaching around 54% penetration by 2030, driven by its heavy reliance on imported oil and gas as well as increasingly stringent carbon regulations. China is set to remain the largest NEV market, with penetration forecast to climb from 52% in 2025 to 85% by 2030, even as subsidies gradually phase out.

Despite surging cost inflation from raw materials (i.e. lithium, copper and ali) that might put downward pressure on the IRR of ESS projects YTD, bringing uncertainties on the progress of ESS projects, we still see secular demand growth of global energy storage market (namely ESS) in the medium term. Growth is set against the backdrop of the improving economics with the lowest-ever prices, LT carbon commitment, and stable power market. China, the U.S, and Europe should hold the commanding lead in global ESS development over the next couple of years, taking up 45%, 14%, and 12% of global ESS installation by 2030 respectively, per our forecasts.

Figure 24. Citi long-run demand assumptions for ESS by region

	2022	2023	2024	2025E	2026E	2027E	2028E	2029E	2030E
ESS Battery Demand	**85**	**166**	**293**	**616**	**895**	**1,207**	**1,470**	**1,734**	**2,093**
YoY		95%	77%	110%	45%	35%	22%	18%	21%
China	30	81	159	246	370	521	637	782	940
US	21	35	60	114	108	140	170	219	300
Europe	21	25	33	102	158	190	214	225	244
Rest of world	12	26	40	156	259	357	449	507	610

Source: Citi Research, ICCSINO

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U.S. Alternative & Renewable Energy
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3. Fuel cells and storage

Fuel cells are electrochemical systems that convert fuels such as natural gas or hydrogen directly into electricity without combustion, offering high efficiency and lower local emissions relative to traditional thermal generation. In stationary and distributed applications, solid oxide fuel cells (SOFCs) are one of the primary technologies deployed, capable of internally reforming natural gas and operating as modular, on-site generation with minimal water use. These characteristics make fuel cells increasingly relevant for data centers, given rising power demand, grid constraints, and the need for reliable, always-on energy. Within this landscape, companies like Bloom Energy (BE) have emerged as leading providers, supporting growing investor interest in fuel cells as a scalable power solution.

Key advantages

Fuel cells offer a compelling set of advantages for mission-critical, distributed power applications, particularly in AI data centers. They provide very low local emissions (minimal NOx/SOx), minimal water usage, and simpler permitting, while enabling fast time to power, with companies like Bloom Energy indicating ~100MW can be delivered in ~120 days and rapidly commissioned on site. Architecturally, fuel cells natively produce DC power, reducing conversion steps and eliminating reliance on transformers, switchgear, and UPS systems, while Bloom Energy's solution supports modern 800V architectures and avoids HV/MV equipment bottlenecks. They also offer attractive, predictable variable costs post-installation, alongside compact footprints (~15 acres per 1GW vs. ~40 acres for CCGT), with BE systems demonstrating ~5.5 year median stack lifetimes, which is expected to increase by two years with advancement in technology.

Capital cost comparison

New greenfield CCGT projects are estimated to have overnight costs of approximately $3,000/kW, according to the Electric Power Research Institute, which we have used in our model. We note this is above the high-end of $1,600/kW per Lazard's June 2025 LCOE analysis, partially as the cost of key non-turbine equipment has continued to increase.

For fuel cells, we assume an upfront capital cost of $5,000/kW and a 30% ITC but no ITC adders. For context, EIA is the only reliable third-party data provider we've encountered with an overnight capital cost estimate for fuel cells. This came in at $7,818/kW, which appears high, especially compared to our modeled ASP for BE of ~$3,050/kW in 2026.

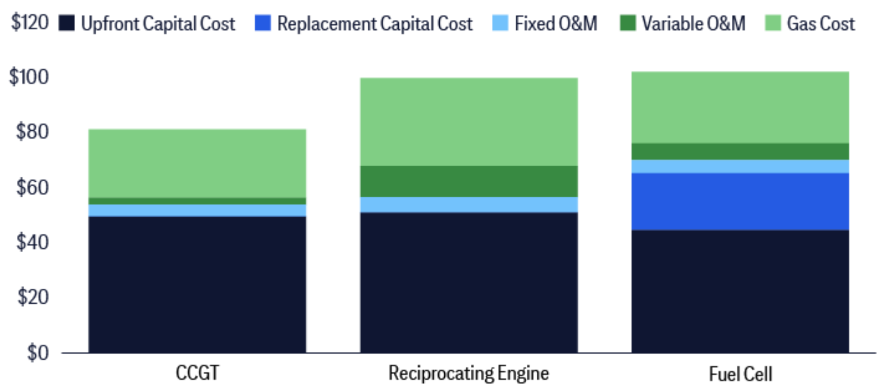
Lastly, we assume stack replacement once every five years. In 2025, BE mentioned a 5.5-year median life before fuel cell replacement. We assume that stack replacement cost is 30% of the total, which equates to ~$1,000/kW. We recognize the replacement cost is a bit conservative and actual cost could be slightly higher.

Variable costs

The next largest cost line item is natural gas. Each technology's heat rate is the primary determinant of natural gas consumed. At a flatlined $4.0 gas cost ($/MMBtu), the gas cost makes up 31%, 32%, and 25%, respectively, for CCGT, recips, and fuel cells LCOE.

Outside of gas, we assume variable O&M costs/MWh of $2 for CCGT, $9 for recips, and $5 for fuel cells.

Figure 25. Comparative LCOE breakdown by technology



Source: Citi Research, Company Reports

Risks to growth

The debate around scaling constraints and supply-chain dependencies centers around availability of scandium, which is a key element used in BE SOFCs. Scandium is largely produced in China and Russia as a byproduct of other mining streams. The location and method of production expose the scandium supply chain to various risks. Especially if SOFC production is scaled rapidly.

Competition

Competition in the SOFC space is increasing rapidly from players such as Ceres Power (and its licensees), Weichai, Doosan, and Delta, some targeting modest pricing advantages but with slower, more capital-intensive ramps.

Figure 26. Fuel cell competitive landscape by technology & scale

Company	Technology	Scale
Bloom Energy	SOFC	Utility/Commercial
GE Vernova	SOFC	Commercial/Utility
Doosan Fuel Cell	PACS » SOFC	Commercial/Utility
Ceres Power	SOFC (licensor)	IP only
Toshiba Energy Systems	SOFC/SOEC	Commercial/Utility
Elcogen	SOFC/SOEC	Component
Mitsubishi Power	SOFC + hybridGT	Utility
Panasonic	SOFC/PEM	Residential
Kyocera	SOFC	Residential/Small
Sunfire	SOFC/SOEC	Industrial
Aisin	SOFC	Residential
Special Power Sources	SOFC	Small distributed
Watt Fuel Cell	SOFC	Small Commercial/Distributed
Convion	SOFC	Commercial/Distributed
Topsoe	SOEC/SOFC	Industrial/Utility
Morimura	SOFC	Small Commercial/Residential
Weichai Power	SOFC (Ceres)	Commercial/Distributed
Petra Power	SOF	Small distributed
Thermax	SOEC (Ceres)	Industrial/Utility
Adelan	SOFC	Small distributed
Delta Electronics	SOFC (Ceres)	Commercial

Source: Citi Research, Company Reports

While SOFC economics are improving, CCGTs remain structurally more cost competitive, leaving fuel cells reliant on niche advantages such as speed-to-power and grid constraint. Current expectations embed meaningful growth (~7GW of installs by 2030), suggesting limited upside for individual players if competition intensifies or adoption falls short.

4. Nuclear

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Vladimir Bystricky
U.S. Industrials and Conglomerates
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In addition to the technologies noted above, we note growing focus on and interest in nuclear energy across both commercial-utility-scale nuclear and small modular reactors (SMRs) with companies such as GEV and MIR poised to play key roles as investment ramps. With respect to commercial-utility-scale nuclear, domestic new build activity remains slow to develop, but we view an increasingly supportive policy environment as encouraging, and planned nuclear plant life extensions and/or restarts we think are indicative of renewed focus on the importance of utility-scale nuclear in meeting growing power demand.

Similarly, internationally we view active and planned investments in commercial-utility-scale nuclear in geographies such as the United Kingdom, France, and Poland as highlighting increased focus on commercial-utility-scale nuclear technology's ongoing role in meeting growing electricity demand.

For SMRs, though still "earlier stage" technology vs. other power equipment mentioned, we think continue to progress toward more significant commercial development over the next decade with an increasingly encouraging policy environment, energy security/reliability, and de-carbonization goals all supportive of continued development and adoption of the technology over time.

Natural gas infrastructure: Why the U.S. needs more

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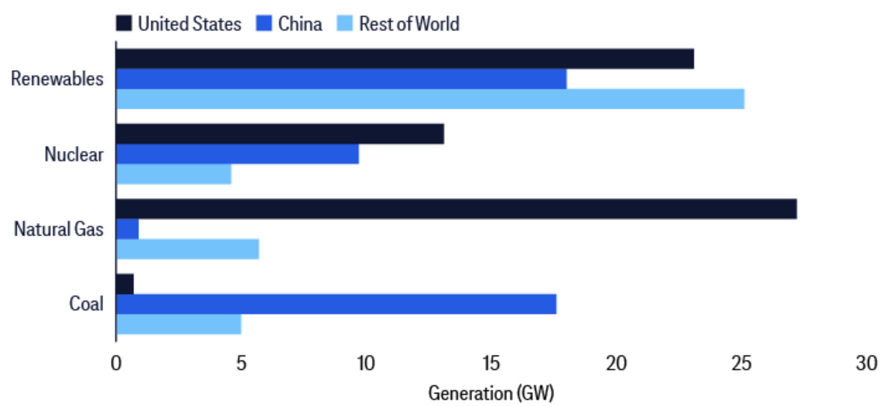
Maggie Lin
Maximilian Layton
Commodities Strategy

Fuel availability and pipelines can become the next bottleneck, shifting constraints from somewhere along the power infrastructure to natural gas infrastructure. Bottlenecks appear more often at the local lateral or facility level, rather than at the interstate pipeline level, where capacity is generally more ample given the need to accommodate peak winter demand for gas from multiple cities. Constraints also include the need to deal with permitting and competition for fuel during cold winter days or hot summer days. Nonetheless, permitting and build time for new laterals and compressor upgrades can at times be months faster than permitting for electric transmissions.

Natural gas as a power solution is globally relevant, but much more competitive in the U.S. U.S. natural gas prices are almost always much lower than other markets in Asia and Europe

The U.S. benefits from having low-cost, vast natural gas fields from shale rock, and associated natural gas production from oil production that also come from shale rock formations. U.S. natural gas prices at the benchmark Henry Hub are generally within a range of $2 to $4/MMBtu.

Figure 27. Potential electricity supply types to power data centers in different regions



Source: Citi Research, IEA

In contrast, many countries in Asia and Europe need to import more expensive natural gas, especially LNG. Asian LNG and European natural gas prices did fall to around $8/MMBtu or slightly lower in recent years but surged to ~$20/MMBtu or higher, notwithstanding the mega price surge in 2022 during the Russia-Ukraine war. Although the global LNG glut expected towards the end of this decade should bring natural gas prices in Asia and Europe down to the $5 to $8/MMBtu range, it is not a certainty. These levels are still higher than U.S. natural gas prices. By definition, natural gas prices in countries that need to import LNG should be higher than U.S. prices, since the U.S. is one of the largest LNG exporters globally. Thus, delivered prices of U.S. LNG have to be U.S. prices plus costs of LNG liquefaction, shipping, and regasification.

Equity Views

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Commercial momentum still accelerating

Power demand and data center growth have spurred a new wave of natural gas infrastructure development. Our natural gas focused Midstream coverage sanctioned ~$38bln of projects tied to power demand growth over the past three years which has led to capex budgets expanding to record levels. We forecast annual capex over the next five years to be >2x the prior five years as a result of this sudden demand shift for natural gas infrastructure. Specifically, we estimate annual growth capex of >$17bln across five companies over the next five years -though power gen only represents a piece of the overall capex estimates.

We anticipate most of this power gen capex to be allocated to the U.S. Midwest and Southeast. Nearly all WMB's BTM projects are located in Ohio while several companies have announced large pipeline projects in the region. Notably, DTM also points to ~7.5bcf/d of incremental gas demand in the Midwest from Utilities. In the Southeast, WMB and KMI sanctioned a combined >6bcf/d of gas pipe expansion in the Southeast.

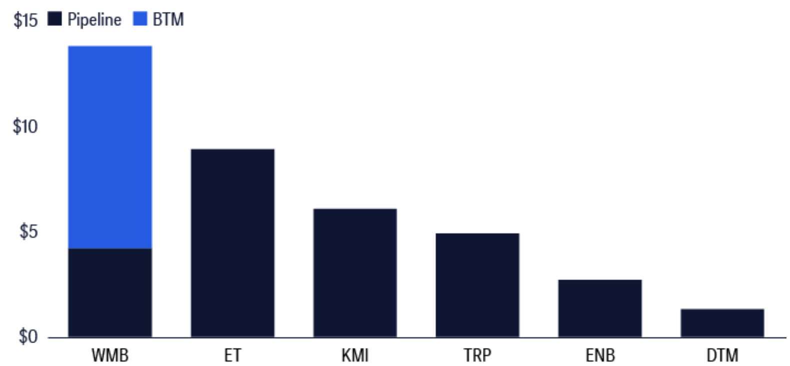
The commercial pace appears to accelerate and thus far has surprised to the upside. Thus, balance sheets are being stretched to fund this capital and could necessitate external financing to fund a portion of the capex. WMB is open to financial partners to support partial funding of its >$20bln Power Solutions opportunity set. TRP may expand its annual capex spending limit once cash flows arrive in 2027+ to support higher leverage. KMI may be in a position to increase growth capex to over $4bln given the size of its unsanctioned backlog, up from $3+bln currently.

We still appear to be in early stages of project development for two reasons. First, our U.S. Utilities team forecasts 96GW of net natural gas generation needed to support power demand growth, where data centers represent 1/3 of total U.S. power demand growth. Such power generation represents ~15-bcf/d of incremental natural gas. Second, commentary indicates gas Midstream operators have large "shadow backlogs" that are, in some cases, multiples of the sanctioned project backlog.

We expect most of the capex to be spent on transmission, specifically on brownfield expansions.

WMB may be the exception with >$9bln of behind-the-meter projects sanctioned and a 6 GW incremental opportunity set. To the extent we see a shift to "bring-you-own generation" or BYOG we believe WMB has the ability to pivot to larger scale CCGT power generation which may already be included in its 6GW opportunity set.

Figure 28. Sanctioned midstream power demand capex ($ billion)



Source: Citi Research, Company Reports

The next phase of growth may encounter some pushback

While the opportunity set is robust, growing data center NIMBYism and energy affordability concerns are driving bipartisan pushback to data center expansion across many states.

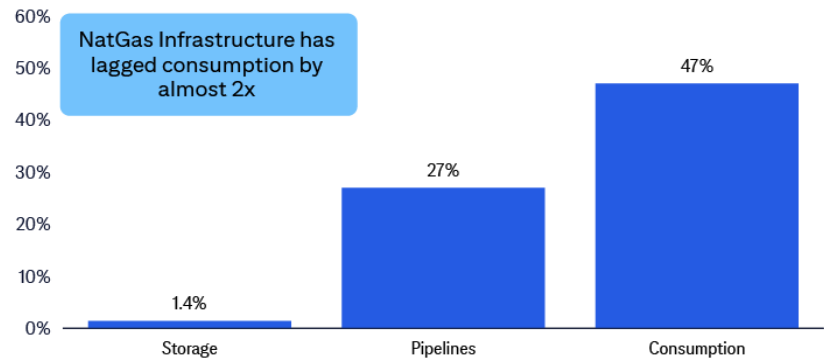
Specifically, local moratoriums on data center development across several states along with broader state moratorium proposals are beginning to emerge. The rising ratepayer concerns could mean developers will increasingly need to "bring your own power" to support data center demand. **13 states have introduced state-wide legislation to place temporary moratoriums on data center development – though two states tabled or vetoed bills (ME, SD)**. All active bills appear to be in early stages of the legislative process. **The bipartisan pushback comes during a midterm election cycle with 36 gubernatorial elections in 2026**. Power and energy affordability are becoming a growing topic during this election season as a Politico poll (Feb 17) found nearly half of Americans expect data center energy costs to be a campaign issue. The PJM power market is seeing power demand growth outpace supply buildout which could lead to capacity shortfalls and elevated rates. The load concentration also created tension among states on cost allocation for new transmission additions.

We're here because of years of underinvestment

Rising natural gas demand and pipeline underinvestment have created the most attractive commercial environment in recent memory, with U.S. natural gas pipeline capacity effectively full. We analyzed the top-50 pipelines in the U.S. which move ~90% of natural gas. During peak periods, those pipelines are over 90% full and, in some cases, temporarily breach their operating capacity.

Due to years of regulatory hurdles, new natural gas infrastructure has been stymied, which has led to natural gas demand outpacing infrastructure by nearly 2x. Storage is even more extreme and now seeing a renewed push for development. We estimate as much as 35 bcf/d of new pipelines would need to be built in order to return to historical levels of pipeline redundancy and capacity.

Figure 29. Demand has far outpaced natural gas infrastructure



Source: Citi Research, EIA

This underinvestment has now placed midstream companies in a strong negotiating position. Notably, multiple recent large pipeline open seasons have been 2-3x oversubscribed vs. the initial capacity (DTM/TRP).

The tight market also manifests in stronger project return. Pipeline projects are broadly being commercialized at ~5-6x build multiples due to the favorable market dynamics and capital efficient, brownfield nature of expansions. We believe these economics likely persist given the favorable demand growth environment and limited excess supply.

Regional power market views: U.S., Europe, China, Australia

The global surge in AI adoption is reshaping energy landscapes, with distinct regional approaches and challenges emerging in the power markets of the U.S., Europe, China and Australia. Their unique regulatory frameworks, existing energy infrastructure, and strategic priorities drive different development pathways.

Ryan Levine
Xinru Yin
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U.S. Utilities
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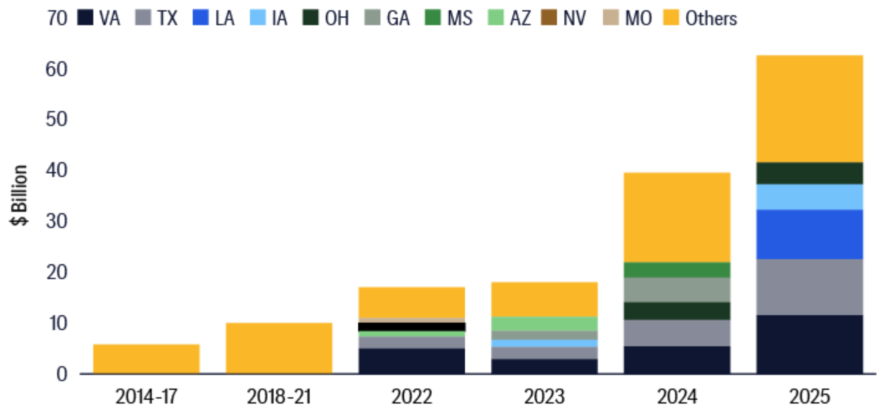
1. U.S. market challenges: Some uncertain regulations and labor as additional concerns

Different regions in the U.S. are competing to add data center driven load and new generation to power the load by working through their unique regulatory approval processes. These processes are moving at an uneven pace and are increasingly becoming a gating item for load growth and utility capex timing. Some regions are leading the charge with expedited processes. Midwest (MISO) and Southeast (SPP) regions have created expedited pathways for generation to come online near-term to help with reliability. Texas (ERCOT) is shifting toward a large load batch study framework, studying data center project proposals in batches rather than individually to speed up transmission buildout. Meanwhile, other regions are dealing with limited grid capacity, problems with power market incentives, and external factors such as politics and weather which create timeline uncertainty. The Mid-Atlantic (PJM) is making a new set of rules to incentivize new generation buildout (Reliability Backstop Procurement) and to regulate data center load to make sure other customers still get power during peak usage (Connect & Manage). California is revising load forecast methodologies and utilities are doing cluster studies of data center load, but the opportunity set is limited by externalities including wildfire politics and access to capital.

Figure 30. Data center construction spend ($ billion)



Source: Citi Research, Dodge Construction, Federal Reserve Bank of Dallas

This uneven pace of regulatory processes around the country creates a lack of clarity around when data center load and/or matching generation will come online, which could also lead to timeline mismatch and uncertainty around transmission buildout that would connect data centers to the grid.

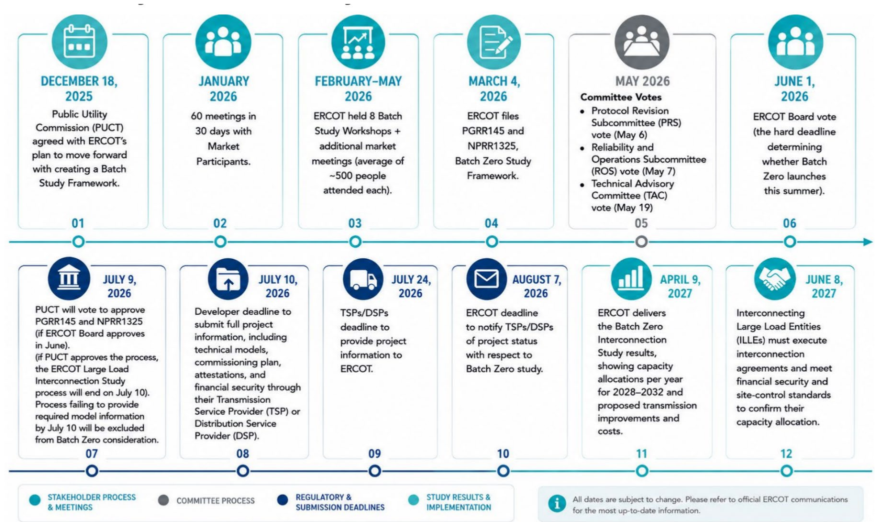
On a local level, each county within the states has their unique data center politics that is complicating state and regional policy making. In some regions, the key issue is utility bill affordability, while in other regions, the key issues are air permits, water usage, noise, property values, and general views of AI.

While each region faces different sets of political/regulatory challenges, this contest for speed to power creates opportunity for electric wires and generation investments.

ERCOT: Texas

Large load approval moves from ad hoc review to batch study: ERCOT's approval bottleneck is increasing on large load interconnections, largely data centers, rather than only new generation. Loads of 75MW+ must go through the large load interconnection process, and ERCOT has developed a batch study framework to assess multiple large load proposals together (selection expected in August). This should improve visibility for Transmission Service Providers (TSPs) and utilities, but also raises bar for project readiness, commissioning, financial commitment, and transmission impact analysis.

Figure 31. ERCOT: Batch Study Framework Key Milestones

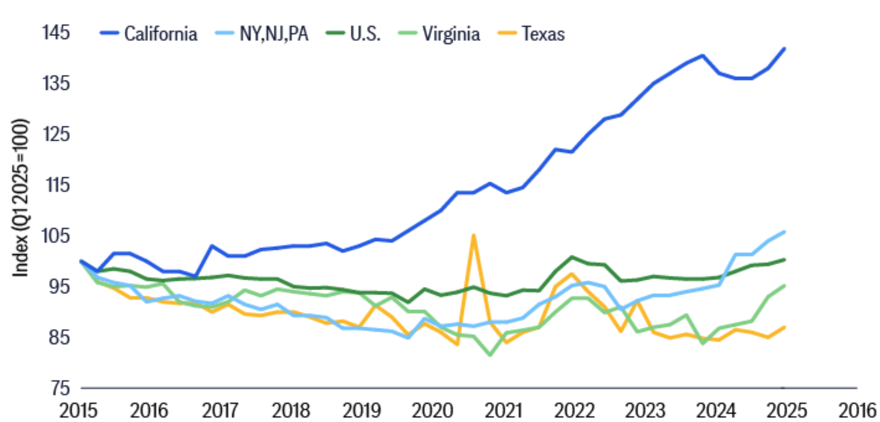


Source: Citi Research, ERCOT

For ERCOT, the issue with their stakeholders isn't utility bill affordability, unlike many other regions in the U.S. with data center growth prospects.

There are over 300GW of projects requesting grid access via the ERCOT batch study process, but the grid can't handle all this load by the 2032 batch study in service date. During our recent visit to Texas to meet ERCOT and other stakeholders, ERCOT emphasized that the transmission infrastructure can only handle ~150 to 160GW of load compared to ~95GW current utilization before any potential transmission expansion. The regulators also need to prepare for organic load growth of 2-2.5% as the Texas economy continues to grow.

Figure 32. Utility bills rising in many states but less so in Texas

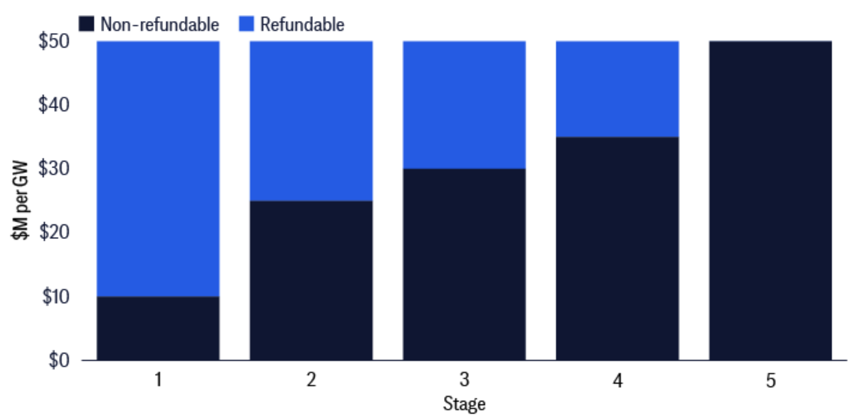


Source: Citi Research, EIA, Federal Reserve Bank of Dallas

This constraint is forcing Texas to be selective on which load it will allow to connect to the grid based on 1) transmission limitations, 2) site location, 3) neighboring load, and 4) load shaping, along with anticipated grid expansion plans. Beyond grid connection, we expect the Texas grid operators to lean heavily on two policy tools – WLPUN, which allows load to build back up generation to provide it extra power without utilizing the grid subject to several restrictions, and PCLR, which allows data centers and large load customers to access the grid more hours a day during non-peak time periods. These policy tools should allow data centers to ramp their grid access in a managed way while giving them options to supplement their load ramps with backup generation under certain conditions.

While we view Texas to be a very investment friendly state regarding data centers, there are several practical challenges with no clear solution. On the planned new 765kV transmission line buildout, strong landowner opposition could make targeted timelines and cost estimates hard to maintain. We expect these lines to not meet 180-day approval timelines, have some cost overrun, and potentially get rerouted with increased cancellation risk. Air permits are becoming a bigger political issue, particularly in rural communities as backup generation runs 24/7 compared to large-scale gas generation permits which allow ~40% utilization. In addition, there are physical limitations to the gas pipeline network which limits availability of gas generation feedstock. This dynamic effectively puts power generation in competition with LNG. On water usage, data centers may need to commit to building some reservoirs and closed-loop DC systems to offset water impacts from data centers and new power plants. Lastly, there is a risk that data center developers with projects in-flight will not complete their buildout given only 20% of the $50,000 per MW required down payment is non-refundable.

Figure 33. Citi's view on required down payment from data centers: Refundable vs. non-refundable at various stages



Source: Citi Research

PJM: Mid-Atlantic

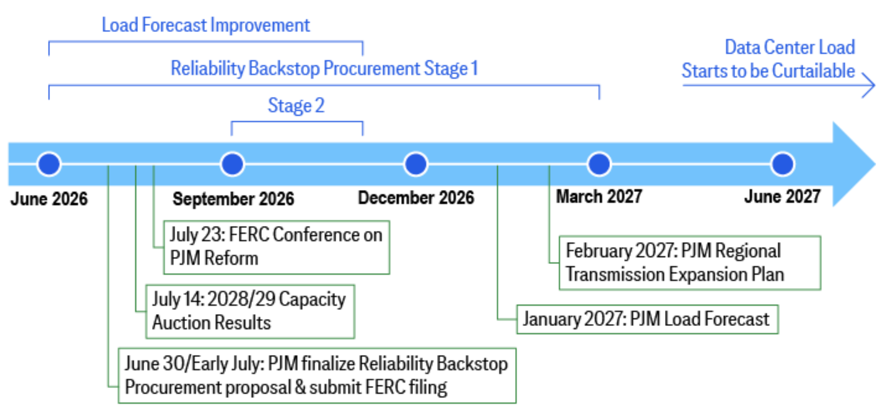
Reliability backstop procurement attempting to incentivize new generation near-term: The Mid-Atlantic region is faced with unprecedented data center load growth with a large established market in Virginia, and emerging clusters in Pennsylvania, Ohio, etc. However, the high cost of building new generation in this region along with structural problems with the current market mechanism have failed to incentivize power developers to build new generation in the near-term to power the projected load addition. This is at the top of mind of the federal government as well as state governments, who are pressuring for an expedited process to solve this problem.

The grid operator PJM is setting new rules on a special two-staged process to procure generation (reliability backstop procurement). As of now, stage 1 of process is set to begin in June and end in March 2027 (bilateral contracting), and stage 2 will run from September to November 2026 (centralized procurement). This process is already relatively fast-paced compared to history, despite the inherent complexity of the issue and the lengthy back-and-forth between stakeholders including power developers, utilities (wires companies), data centers, state governors, and others. We think the PJM approach should yield positive outcomes, but is not a complete solution to the problems being faced.

Working through policy challenges: PJM is well aware of the challenges and hired a new CEO who came out writing that "the current situation is not tenable" with a detailed report around potential path forward for the PJM power market. It was emphasized that the region has years, not decades, to make choices deliberately.

Challenges with current reliability proposal: While there is a draft process in place, in our view, the PJM design described above has a lot of challenges. The current policy design causes the regulated wire companies to assume several risks without being compensated. This has caused these grid companies to indicate publicly that the solution isn't workable without state-level legislative reforms (which is hard given the 13 states involved in PJM). On the flip side, some stakeholders would like the load-serving entities (LSE) to have more responsibilities to help solve the problem, but many of them are smaller with less than investment grade ratings which pose other challenges.

Figure 34. Key event catalysts to PJM load discovery process



Source: Citi Research, PJM

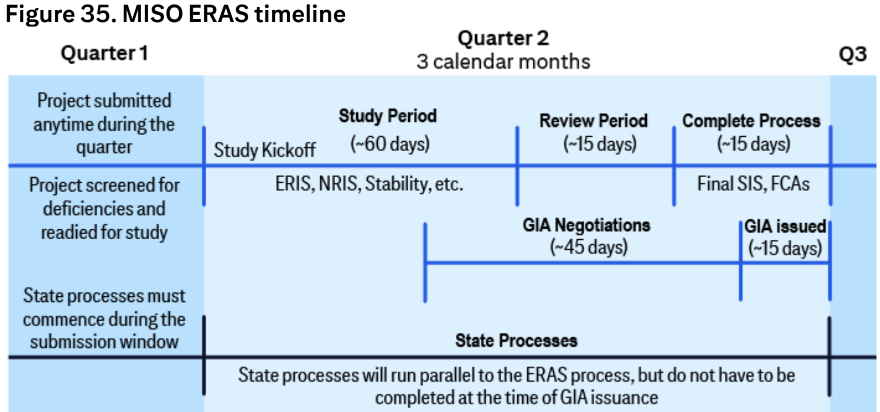
PJM ultimately could work through issues, but timeline uncertain: We believe that PJM along with FERC and state governments will come to a realistic solution that will support data center and power generation growth in the region. As politics around utility bill affordability is likely to receive fewer headlines after the next election cycle, and the stakeholder engagement continues, we are optimistic in the longer-term outlook. One potential solution could involve Federal Energy Regulatory Commission (FERC), which oversees PJM and recently indicated they are interested in reforming PJM's governance and stakeholder process with a conference scheduled for July 23. See our recent notes on PJM for detailed discussion: PJM's Revised RBP Proposal Accelerates Timeline & Address Concerns, A Step to the Right Direction; Alternative Proposals for PJM Backstop Auction Foreshadows Potential Changes; Big Change to PJM Power Market Coming? Landmark Report Points to Potential Structural Changes; PJM Reliability Backstop Procurement Hearings Takes - Expert Event; Takeaways from Expert Call on PJM Reliability Backstop Procurement; PJM Backstop Auction First Take

MISO: Midwest

The Midwest is an attractive data center market, but has some physical constraints such as transmission and gas pipelines, as well as local politics. The region has several advantages such as relatively reasonable grid policy, access to gas, and frameworks for data center growth that several states have created. One positive regulatory policy is the ERAS process that is put in place to help encourage data center development in this region.

ERAS creates a faster but narrow path for reliable gen: MISO's Expedited Resource Addition Study (ERAS) is a temporary process meant to move qualified generation through interconnection faster where projects address resource adequacy or reliability needs. The process is capped with strict eligibility and is not a broad queue-clearing mechanism. For utilities, ERAS can support faster capacity additions tied to large loads as well as retirement replacement; however, the limitation is that only projects with site control, offtake, RERRA verification, and near-term COD readiness qualify.

Figure 35. MISO ERAS timeline



Source: Citi Research, MISO

CAISO: California

Data centers are not a significant driver of load growth. The conversion from potential interest, to pipeline, to final engineering stages, and to construction remains limited vs. other regions. Grid constraints, high cost of grid upgrades, overhang from wildfire risks and resulting reduced ability for utilities to access capital to execute on these growth opportunities pose uncertainty to data center growth.

Other regions

SPP, with coverage of part of the U.S. Midcontinent, received approval from FERC for its High Impact Large Load (HILL) process in Jan'26, but market participants indicated that SPP's ERAS process is most helpful. SPP as a fully regulated market took action relatively quickly and established the ERAS process, which is designed to encourage new generation that can service incremental power demand for the region. During our recent meetings with SPP, they highlighted several other initiatives to help attract large load. SPP may be relatively attractive for BYOG, as it has a defined pathway for large loads paired with generation.

NYISO, covering New York state, faces the usual load-study issues, as well as tight downstate capacity, transmission constraints, local

permitting, and clean-energy/resource adequacy constraints. Data centers upstate may be more feasible than downstate.

The Pacific Northwest has plenty of hydro and low carbon emissions overall, but BPA (Bonneville Power Administration) transmission access is scarce and queue reform matters (BPA, April 30).

The Southeast is generally not part of any ISOs, so processes and progress are done at the regulated utility level and on a state-by-state basis. Utilities handle large loads mostly through vertically integrated resource planning, special contracts, and state commission approvals. The Southeast can move faster than many other markets. Georgia, with a lower carbon power resource mix, reports more data center requests.

2. Europe's AI power opportunity

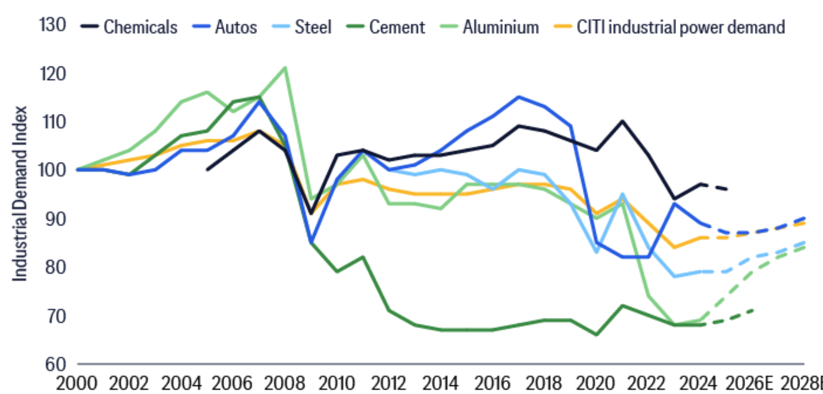
Jenny Ping
Chris McDonagh
European Utilities and Renewables Equity Research

Structural system differences: fuel, policy and cost

Europe's data centre opportunity sits within a structurally different power system to the U.S., defined by limited access to low-cost domestic fossil fuels and stronger policy commitment to decarbonisation. This results in a system anchored in renewable generation, higher power costs, and constrained availability of dispatchable capacity.

At the same time, underlying power demand has been weak historically (EU27+UK demand down ~10% over the past 15 years, with industrial demand down ~20%). Reserve margins remain broadly comfortable, with baseload thermal utilisation trending lower. Against this backdrop, commodity pricing – particularly gas – remains the key marginal driver of power prices, despite calls across Europe to de-link power and gas prices.

Figure 36. Europe has de-industrialized, creating spare capacity power generation

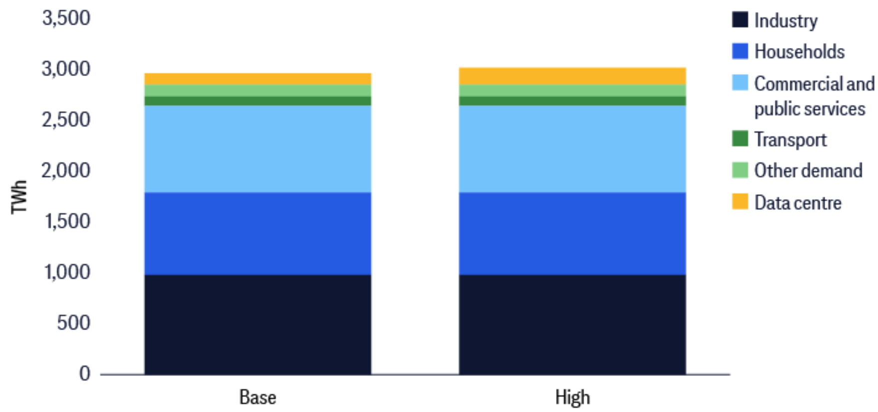


Source: Citi Research Estimates

Data centre growth part of electrification investment cycle

Europe is undergoing an investment cycle to support electrification in transport, heating, and industry demand. Data centres account for ~2% of current demand, with a further 2-3% from projects under construction or latent capacity, but the forward trajectory remains uncertain, with estimates broadly in the region of 1-3 times growth by 2030.

Figure 37. Even after 3x growth, data centre demand accounts for only ~6% of the total



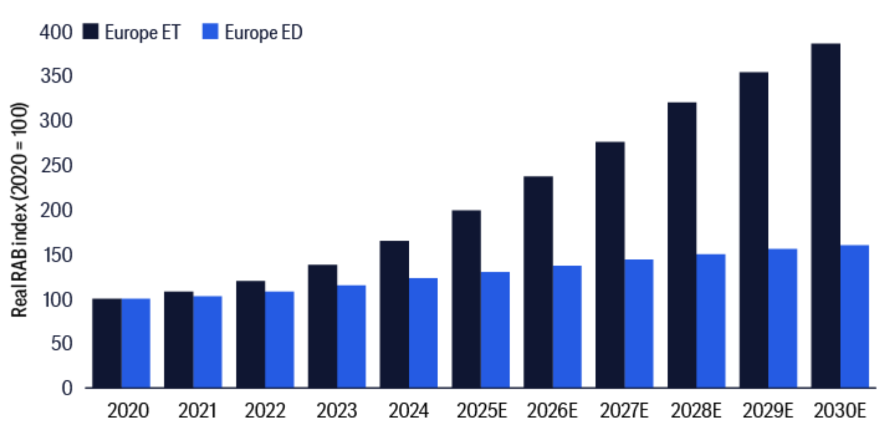
Source: Citi Research, ENTSOE, DUKES

Structural constraints – long connection queues, renewable intermittency, and high energy costs – are not easily resolved, meaning a portion of announced capacity risks remaining "bragawatts" rather than realised load. In this context, we see data centres as a marginal accelerator rather than a primary driver of system growth, with their ultimate contribution dictated by infrastructure readiness.

Grid capex is the core investment opportunity

The key implication is that grid expansion is the primary beneficiary of this dynamic. Electrification in a distributed renewable generation system requires significant reinforcement of transmission and distribution networks. Even under slower data center growth and AI adoption scenarios, grid capex remains supported by relatively stable regulated frameworks and broader system needs.

Figure 38. Power transmission and distribution asset base evolution in Europe



Source: Citi Research Estimates, Company Reports

Power ready land unlocking value

Within a constrained grid environment, value is increasingly concentrated in assets that offer immediate or accelerated access to power. UK transactions have demonstrated that decommissioned coal sites – previously assumed to be worth nothing – can command significant premiums due to embedded grid connections and permitting advantages.

Figure 39. Site sale examples

Asset	Location	Seller	Buyer	Timing	Type	Size (Acres)	Value	Cur.
Blyth	UK	Northum-berland Council	Blackstone QTS	2024	Old Coal	235	130	£m
Skelton Grange	UK	Harworth	Microsoft	2024	Old Coal	48	107	£m
Didcot A	UK	RWE	Undisclosed Hyperscaler	2025	Old Coal	—	225	€m
Cottam	UK	EDF	Holtec, EDF & Tritax JV	2025	Old Coal	900	—	—
Lübbenau	Germany	—	Schwarz / StackIT	2025	Old Coal	32	—	—
Montereau-Vallée-de-la-Seine	France	EDF	EDF/OpCore JV	2025	Old Coal	49	—	—

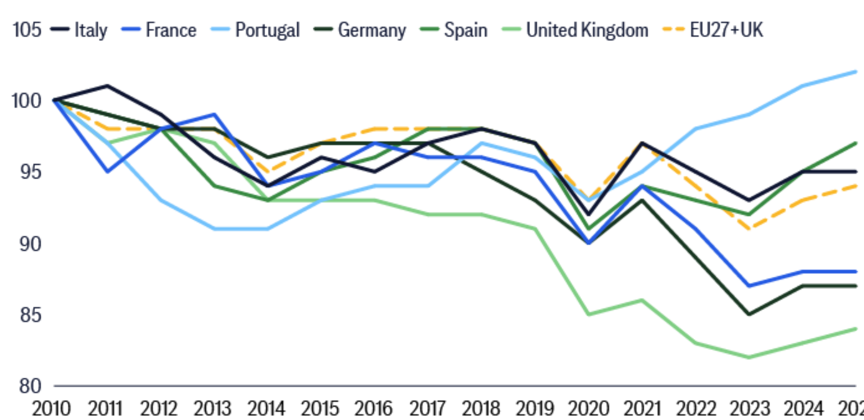
Source: Citi Research, Company Reports

Across Europe, ~83GW of retired coal capacity exists, with theoretically ~50GW suitable for conversion, subject to site-specific constraints. These "powered land" assets effectively bypass long interconnection queues, making speed to power the critical determinant of value. More broadly, control of grid connected sites represents a structural advantage in a system where new connections are scarce and time intensive.

Country-level view – system tightness matters

Exposure to the data center theme is not uniform across Europe and depends on local system's tightness and available capacity. Our bottom-up country analysis of reserve margins shows limited signs of system stress in most markets, suggesting that incremental data center demand is often absorbed within existing headroom rather than driving new build. However, smaller or more supply-constrained systems can see disproportionate impact. Portugal is a notable example, where a relatively small power market combined with large projects such as the Start Campus (0.2 GW under construction, 1.2GW pipeline) development represents a meaningful step change in demand. See our May 2026 report, [EDP (EDP.LS): A Rare Find; Initiate with Buy](), for exposure to this theme.

Figure 40. Portuguese power demand stands out



Source: Citi Research, ENTSOE, DUKES

3. China Power Plant Equipment

Pierre Lau, CFA
Bella Tian
Air Ma
Asian Utilities & Clean Energy
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China pursues an all-of-the-above strategy. Besides solar and wind, China's focus on nuclear and hydro supplements its traditional buildout of fossil-fuel generation.

PRC power plant equipment sector with catalysts from adding more non-coal fired capacity

On nuclear, China approved 31 new units in 2022 to 2024. In April 2025, China approved another 10 new nuclear reactors across five coastal projects with estimated total outlay at Rmb200bn, as part of its ongoing nuclear energy expansion. We expect China nuclear installation volume to grow at a CAGR of 36% over 2025-30E, according to our conversations with CGN (1816.HK, SELL) and CNNP (601985.SS, SELL).

Figure 41. PRC annual new add (ex-replacement) (MW)

Annual new add (ex-replacement) (MW)	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031-60E
Coal	29,400	28,200	47,700	32,490	70,260	50,000	35,000	25,000	25,000	25,000	-40,841
Hydro	23,500	23,900	10,300	13,780	12,150	15,000	15,000	15,000	15,000	15,000	9,233
Natural Gas	7,700	6,300	10,100	18,990	19,920	20,000	20,000	20,000	20,000	20,000	14,546
Nuclear Power	3,400	2,300	1,400	3,930	1,530	3,800	5,000	5,000	6,000	7,000	7,024
Wind Power	47,700	37,600	75,700	79,820	120,480	138,000	150,000	155,000	160,000	165,000	69,733
Solar Power	54,500	87,400	216,000	278,000	317,510	250,000	250,000	250,000	250,000	250,000	118,276
Others	12,900	14,000	7,900	6,200	4,320	4,320	4,320	4,320	4,320	4,320	8,942
Total	**179,100**	**199,700**	**369,100**	**433,210**	**546,170**	**481,120**	**479,320**	**474,320**	**480,320**	**486,320**	**186,913**

Source: Citi Research, National Nuclear Safety Administration

China has achieved 100% domestic production of key nuclear power equipment, and will accelerate self-reliance while continuing to promote international cooperation, especially with major nuclear countries and Belt & Road nations (Reuters, 28 April 2025). It takes around 2-5 years for nuclear equipment makers from receiving new orders to recognize revenue, depending on different equipment. Steam generator and turbine generator production duration are 2-3 years.

On hydro, we project China to add 15GW of new capacity per year in 2026E-30E, driven by more demand for pump storage. We expect China to add new pump storage capacity of 100GW in 2026E-30E, according to the draft outline of the 15th Five-Year Plan (2026-2030). According to "Medium and Long-term Development Plan for Pumped-Storage Hydro (2021-2035)" released by NEA, by 2035, the total installed capacity of pumped-storage hydro will reach approximately 300GW. The mega Yarlung Zangbo Hydropower Project will likely take 15-20 years vs. 12 years for Three Gorges Dam (please see our report: China Electrical Equipment: Takeaway from Expert Call on the Yarlung Zangbo Hydropower Project).

On coal, we expect tendering volume in China to drop to 30-40GW pa in 2026E-30E, from average of 77GW pa in 2022-25, owing to the shifting of coal-fired role from primary power generation to auxiliary peak shaving. China proposed the "three 80GW" policy in 2022 to solve the problem of large-scale power outages in China during the summer of 2021. This policy required to see the start of construction for 80GW of coal-fired power pa in 2022-23, and a total of 80GW put into operation over 2022-24.

4. Australian gas pipeline and utilities

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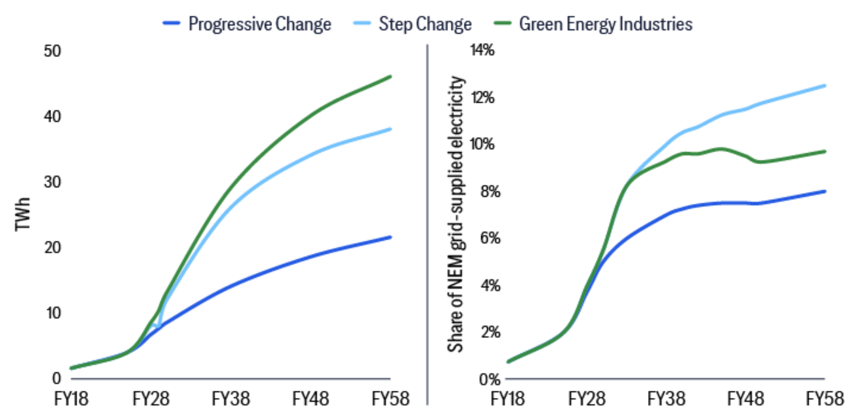
Australia's Energy Market Operator (AEMO) estimates that in FY25 (year ending June 2025), data centers consumed 3.9 terra-watt hours (TWh) of electricity, with 98% of this coming from NEM (East and Central Coast of Australia, and Northern Territory states), and this represented 2% of NEM grid-supplied consumption. **By FY30, data centre energy consumption is forecast by AEMO to expand to 12.0 TWh growing at ~25% CAGR, and is expected to be 6% of NEM grid consumption**. Further growth is expected with energy consumption expected to be 34.5TWh by FY50. Importantly, these numbers reflect the step-change scenario (which we see as a representative base), but these numbers have been revised upward in recent years, highlighting potential for estimates to move towards the more positive scenarios highlighted.

Figure 42. Australia's AEMO forecasts on data center energy consumption, absolute and as share of grid supplied electricity for NEM (East and Central Coast, and NT)



Source: Citi Research, Australia Energy Market Operator (AEMO)

The AEMO has highlighted grid connection requests for 5.4GW of data centre related power requirements, of which 60% are in New South Wales (Sydney and surrounding areas) and the remaining in Victoria (Melbourne). This compares to average energy draw of ~600MW for 1Q 2026.

Increasing challenges lend to potential alternate sources of demand

Similar to other global markets, Australia has its challenges in catering to the booming energy demand, including:

- **Grid stability concerns:** AECO raised concerns on potential grid instability over time with potential for voltage disturbances to cause massive outages. This is being seen as a major concern by the government too.

- **Power pricing impact concerns:** With increased investment needed in power generation and transmission lines to cater to demand for data centers, there have been concerns around power pricing impact from this investment, as companies fund increased investment into the grid.

- **Rising cost of construction:** Higher construction costs is also another key factor, particularly with Australia facing shortages for a range of trades due to a range of public infrastructure work (roads, railways) and work related to the Brisbane 2032 Olympics, in addition to big mining and utility projects underway. This has increased labour costs, and material costs pricing has increased in recent years due to higher inflation and increased supply-chain concerns following the Middle East conflict. Rising costs increase the output costs needed for energy providers to cater to new demand.

Potential behind-the-meter development, with gas the bridging fuel

This expansion in energy demand is positive for all electricity generation providers, and in our view, some of the trends witnessed in the U.S. market could play out here. This lends nicely to the creation of a behind-the-meter power market in Australia. The Australian government is backing natural gas is as a critical bridging fuel for the energy transition ambitions. We therefore see gas as playing a critical role in fulfilling the rising demand for data center power in Australia, in a renewable exposed grid.

Utilities (AGL/ORG) and gas pipeline provider APA benefitting

APA.AX, Buy A$11.10 target price — While early, the tilt towards increasing grid concerns highlights similar movements to those seen in the U.S. market, which has driven the rise of the behind-the-meter power industry. This again could benefit APA via the energy generation business, as well as through potential addition of gas pipelines (like Beetaloo basin), benefitting gas transmission earnings which are >80% of EBITDA; see APA Group (APA.AX) - Secure Energy Infra Income with Upside Potential.

ORG.AX, Buy A$13.00 target price — While the market is increasingly pricing in softer swaps and caps in the NEM, we believe risk is skewed to the upside with renewable development lagging despite government underwriting initiatives. The market is implicitly pricing coal for longer, and we see increasing risk that ORG's Eraring black coal power station will be pushed beyond its current 2029 closure date. This would be supportive of ORG earnings with further risk sharing leverage with the government without sufficient replacement generation in the grid. While DMO8 headwinds will drag on retail tariffs into FY27, we are more constructive on margins into FY28 with the expectation that forward curves re-rate higher through the year. We think ORG's gas peaker fleet is undervalued in the market, particularly as coal exists the grid.

AGL.AX, Buy A$11.50 target price — AGL's transformation into renewable generation and BESS development is lagging with market signals implying volatility will moderate with 49GW of early stage BESS assets connecting to the grid. We think the mid -cycle transition of coal exists should see volatility returning and think BESS assets being commissioning will exceed IRRs returns and benefit from early mover advantages. Likewise to ORG, DMO headwinds will compress retail margins into FY27; however we think the finely balanced market means weather/fleet reliability factors see curve re-pricing risk to the upside.

Actionable ideas

Given the power market evolution and high demand for infrastructure and generation equipment, we are positive on several strategic pathways:

On U.S. Utilities, we are positive on Midwest regulated utilities such as Evergy Inc (EVRG.N, BUY) and PJM electric transmission companies First Energy Corp (FE.N, BUY) and Exelon Corp (EXC.N, BUY) on the evolving policy outlook tied to data centers

On EU Utilities, we recently initiated with a Buy on EDP (EDP.LS, BUY), driven by its growth strategy including EDPR's repowering, Portugal's data center demand, and Brazil's reforms, supported by earnings quality and undervalued position.

On Australia, we are positive on gas pipe provider APA Group (APA.AX, BUY), driven by potential gas pipe additions, and utilities AGL (AGL.AX, BUY) and ORG (ORG.AX, BUY). AGL is poised for strong returns from BESS assets as coal exits. ORG benefits from extended coal plant operation and undervalued gas peaker fleet.

On Asian Utilities and Equipment, we are constructive, driven by robust global demand for high voltage transformers and switchgears to support AC data centers; our top picks include Korean firms Hyosung Heavy (298040 KS, BUY) and LS Electric (010120.KS, BUY) for strong U.S. orders, Chinese peers Sieyuan Electric (002028.SZ, BUY) and TBEA (600089.SS, BUY) for their competitive cost structure and non-U.S. market share, and PRC power equipment maker Dongfang Electric (1072 HK, BUY) benefiting from nuclear, gas, and hydropower equipment demand.

On U.S. Machinery, we are bullish on Caterpillar (CAT.N, BUY) and Cummins (CMI.N, BUY) due to strong data center revenue forecasts, with CAT projected to reach ~$20bn by 2030 and CMI ~$12.5bn, on capacity expansion and market positioning.

On Batteries and Energy Storage, we are bullish on CATL (300750.SZ, BUY), EVE Energy (300014.SZ, BUY), CALB (3931.HK, BUY) and Ganfeng Lithium (1772.HK, BUY), driven by government pushes for renewables and electrification due to energy security concerns, global net-zero goals, rising NEV adoption, and long term ESS demand.

On U.S. Midstream, we are bullish on The Williams Companies (WMB.N, BUY) and DT Midstream Inc (DTM.N, BUY) due to WMB's sanctioned gas pipe expansion and strategic financing for power solutions, and DTM's identified incremental gas demand in the Midwest, propelled by further power demand and data center growth.

Key Stocks Referenced:

AGL Energy Ltd (AGL.AX; A$8.38; 1; 25 Jun 26; 16:00) | APA Group (APA.AX; A$10.63; 1; 25 Jun 26; 16:00) | CALB Group Co Ltd (3931.HK; HK$24.88; 1; 24 Jun 26; 16:10) | Caterpillar Inc (CAT.N; US$994.45; 1; 24 Jun 26; 16:00) | CATL (300750.SZ; Rmb395.36; 1; 24 Jun 26; 15:00) | CGN Power (1816.HK; HK$2.8; 3; 24 Jun 26; 16:10) | China National Nuclear Power (601985.SS; Rmb8.98; 3; 25 Jun 26; 15:00) | Cummins Inc (CMI.N; US$694.87; 1; 24 Jun 26; 16:00) | Dongfang Electric (1072.HK; HK$23.34; 1; 24 Jun 26; 16:10) | DT Midstream Inc (DTM.N; US$147.09; 1; 24 Jun 26; 16:00) | EDP (EDP.LS; €4.38; 1; 24 Jun 26; 16:30) | Eve Energy (300014.SZ; Rmb66.81; 1; 25 Jun 26; 15:00) | Evergy, Inc. (EVRG.N; US$85.81; 1; 24 Jun 26; 16:00) | Exelon Corp (EXC.O; US$46.9; 1; 24 Jun 26; 16:00) | FirstEnergy Corp (FE.N; US$47.82; 1; 24 Jun 26; 16:00) | Ganfeng Lithium (1772.HK; HK$56.05; 1; 24 Jun 26; 16:10) | Hyosung Heavy Industries (298040.KS; W3280000.0; 1; 25 Jun 26; 15:45) | LS Electric (010120.KS; W219000.0; 1; 25 Jun 26; 15:45) | Origin Energy Limited (ORG.AX; A$10.86; 1; 25 Jun 26; 16:00) | Sieyuan Electric (002028.SZ; Rmb188.5; 1; 25 Jun 26; 15:00) | TBEA Co (600089.SS; Rmb23.2; 1; 25 Jun 26; 15:00) | The Williams Companies Inc (WMB.N; US$75.87; 1; 24 Jun 26; 16:00)

A Conversation with Citi Wealth

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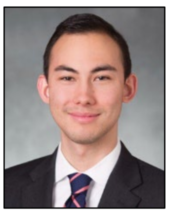
***JP Coviello, Head of Portfolio Strategy,** joined Citi Wealth in August 2025 and is responsible for leading global portfolio and investment insights within the Chief Investment Office. He also serves as a member of Citi Wealth's Global Investment Council, responsible for establishing tactical asset allocation guidance for client portfolios.*



***Hobey Kuhn is a Macro Portfolio Strategist** within the Chief Investment Office for Citi Wealth. He joined in November 2025 and is responsible for developing differentiated, actionable investment insights across asset classes, providing portfolio guidance, and enhancing the rigor of the team's asset allocation, insights, and risk management frameworks.*

As Power & Energy demand increase globally, where does Citi Wealth see the most compelling investment opportunities for client portfolios?

We see investment in global energy infrastructure as a multi-year structural growth story, driven by the massive capital spending required to support global electrification, the AI-driven data center buildout, and "energy security," as countries look to diversify their energy sources. Three supply-chain crises over the last three years (COVID[8], Russia-Ukraine war[9], Middle East conflict[10]) have highlighted the strategic vulnerabilities associated with dependence on foreign sources or concentrated supply chains for critical economic inputs. Our view is a compelling set of potential opportunities can be found across four key pillars that form the backbone of this transition:

1. **Grid and infrastructure modernization:** This is the foundational layer. We believe the existing grid infrastructure is unprepared for the surge in power demand. The International Energy Agency (IEA) estimates annual grid investment must increase significantly to over $600 billion by 2030 to meet projected needs[11]. The Texas grid operator (ERCOT) has seen its pipeline of projects seeking grid connection increase six-fold in the last 18 months[12]. We see potential opportunities in companies providing the essential hardware, software, and services for grid expansion and modernization.

[8] The COVID-19 pandemic began in late 2019 and the global emergency phase officially ended in May 2023.

[9] The Russia-Ukraine war began in February 22, 2022 and is ongoing.

[10] The Middle East conflict between the U.S. and Iran began in February 28, 2026.

[11] International Energy Agency (IEA), "Electricity Grids and Secure Energy Transitions." June 3, 2023.

[12] Electric Reliability Council of Texas (ERCOT) as of May 2026.

2. **Nuclear power:** After years of being overlooked, nuclear power is re-emerging as a critical source of clean, reliable, 24/7 baseload power. There is bipartisan political support in the U.S. and ambitious targets globally. China, for example, aims to build 150 new reactors by 2035, and the U.S. is providing significant incentives through programs like the Inflation Reduction Act[13] and the Department of Energy's Title 17 loan program[14] to de-risk and encourage new large-scale reactor construction. While past projects have faced significant delays and cost overruns, there is a clear path to cost reduction through standardization and repeatability, as seen in other countries. This suggests that while execution risk remains, the long-term potential is not yet fully priced into the sector.

3. **Renewables:** In our view, renewable energy sources like solar and wind are central to achieving climate goals and enhancing energy independence. The COP28[15] pledge to triple renewable energy capacity by 2030 implies a need for over $1.2 trillion in average annual investment. This creates durable demand for companies across the renewables supply chain.

4. **Battery and storage solutions:** The intermittent nature of renewables makes grid-scale energy storage a non-negotiable component of the energy transition. To support the growth in renewables, the IEA projects the global energy storage market will need to expand six-fold by 2030. This creates a significant opportunity for companies involved in battery technology, manufacturing, and deployment.

The U.S., South Korea, and Taiwan have been key beneficiaries of investor attention, particularly around AI and technology supply-chain themes. Do you recommend clients still add exposure to these markets?

Yes. We believe the structural growth story around AI technology capex and its supply-chain remains intact and continue to recommend exposure, but with a nuanced and selective approach. We are likely in the middle of a multi-year capex cycle related to AI, which will likely continue to benefit stocks in the U.S., South Korea, and Taiwan. However, after a period of exceptionally strong performance and massive investor inflows, equity valuations and positioning must be carefully considered.

South Korea and Taiwan: These equity markets are at the epicenter of the AI hardware buildout, with leading positions in memory (South Korea) and logic chips (Taiwan). Their equity markets have performed phenomenally, driven by staggering earnings growth. However, we are cognizant of potential risks.

- **Valuation:** Near-term price-to-earnings (P/E) ratios may appear reasonable, but this is due to "supranormal" profits that may not be sustainable indefinitely. On other metrics, such as price-to-book value, valuations are at or near all-time highs.

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[13] The Inflation Reduction Act of 2022 is a U.S. federal law aimed at reducing the deficit, lowering prescription drug costs, and investing in clean energy and climate initiatives.

[14] The Department of Energy’s (DOE) Title 17 Loan Program is a federal financing initiative that enables the DOE to guarantee third-party loans for qualifying energy-related products.

[15] Conference of the Parties is an international climate meeting held annually by the United Nations

- **Positioning:** These markets have seen unprecedented fund inflows and a surge in retail investor participation, which could exacerbate volatility in a downturn.
- **Concentration:** The markets are highly concentrated in a few large semiconductor names, making them a focused but potentially high-beta play on the tech cycle.

Our view is to maintain exposure as a medium-term theme, but acknowledge the potential for near-term volatility.

United States: The U.S. offers a more diversified and, in our view, durable way to invest in the theme. The potential opportunity set extends beyond semiconductors to also include:

- The **hyperscaler companies** that are driving the demand for AI infrastructure.

- The entire **power infrastructure value chain**, from utility operators to equipment manufacturers, which may benefit from the domestic grid and power generation buildout.

- A supportive **policy environment**, with bipartisan backing for industrial policy aimed at re-shoring critical supply chains and modernizing energy infrastructure.

In conclusion, while we believe in caution given the recent run-up, we also believe the long-term nature of the AI and electrification investment cycle supports continued exposure to these key markets. We favor a diversified approach, with a particular emphasis on the broad and deep opportunity set within the U.S.

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Appendix A-1

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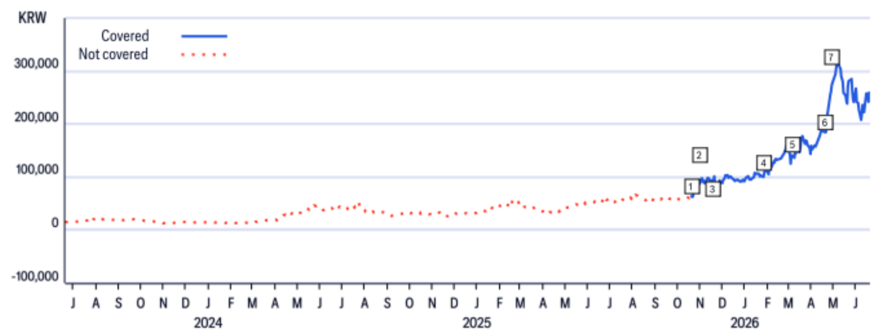
IMPORTANT DISCLOSURES

LS Electric (010120.KS)

Ratings and Target Price History
Fundamental Research

Analyst: Pierre Lau, CFA



	Date	Rating	Target Price	Closing Price
1	20-Oct-25 06:13:00	*1	*76,000.00	63,900.00
2	31-Oct-25 10:56:28	1	*110,000.00	87,000.00
3	18-Nov-25 12:37:11	1	*112,000.00	91,800.00
4	27-Jan-26 10:50:39	1	*130,000.00	107,400.00
5	06-Mar-26 11:35:35	1	*168,000.00	140,800.00
6	21-Apr-26 13:14:46	1	*230,000.00	184,700.00
7	29-Apr-26 06:05:57	1	*320,000.00	273,000.00

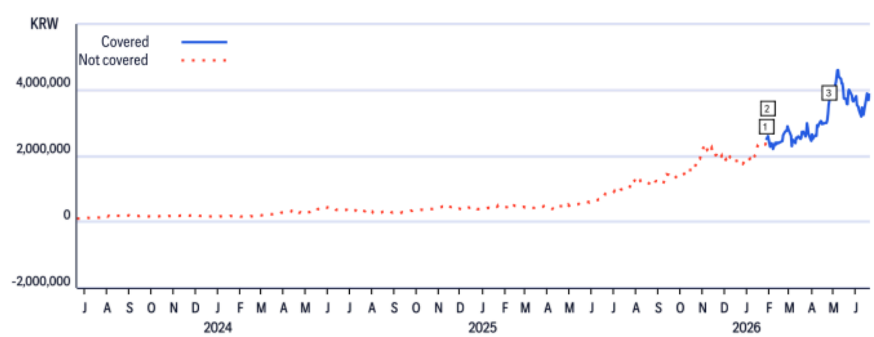
*Indicates Change

Rating/target price changes above reflect Eastern Time

Hyosung Heavy Industries (298040.KS)

Ratings and Target Price History
Fundamental Research

Analyst: Pierre Lau, CFA



	Date	Rating	Target Price	Closing Price
1	28-Jan-26 02:00:37	*1	13,000,000.00	2,518,000.00
2	30-Jan-26 10:48:39	1	13,200,000.00	2,603,000.00
3	26-Apr-26 19:14:53	1	14,500,000.00	3,552,000.00

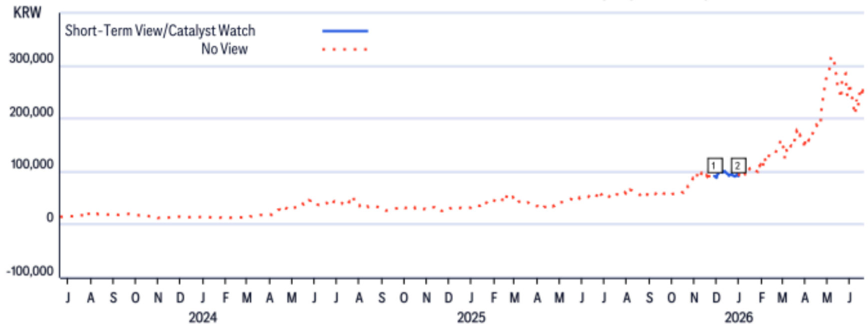
*Indicates Change

Rating/target price changes above reflect Eastern Time

LS Electric (010120.KS)

Short-Term View/Catalyst Watch Research

Analyst: Pierre Lau, CFA



	Date	Action	Expected Direction	Duration	Closing Price
1	30-Nov-25 20:31:17	Add CW	Upside	30 Days	92,000.00

	Date	Action	Expected Direction	Duration	Closing Price
2	31-Dec-25 21:13:37	Remove CW	Upside	30 Days	92,000.00

CW - Catalyst Watch , STV - Short-Term View

Rating/target price changes above reflect Eastern Time

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	12 Month Rating			Catalyst Watch		
Data current as of 01 Apr 2026	**Buy**	**Hold**	**Sell**	**Buy**	**Hold**	**Sell**
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